Exact Coordinate-Balancing Calculus and Sharp Stability for Symmetric Lattice Cross-Covariograms

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Abstract

For finite $A, B \subset \mathbb{Z}^d$, write $g_{A,B}(u) = |A \cap (B - u)|$. We study how this cross-covariogram varies as the coordinate composition of u is balanced along a fixed ℓ_1 shell. A centered exchange-fiber property, satisfied by lattice sections of compact convex sets invariant under signed coordinate permutations, decomposes every unit balancing transfer into parity-fiber endpoint exposures in $\{0, 1\}$, yielding the Schur order on each shell and a fiberwise equality certificate for comparable shifts. For two equal Lee balls of radius t in dimension $d \geq 3$ and comparable shifts on an active shell $2 \leq s \leq 2t$, the overlap deficit is at least a closed constant times the number of unit transfers needed; a central-box section theorem evaluates that constant in lower-dimensional Lee-ball counts. Incomparable shifts may share a value, so equality statements are order-theoretic. For $4 \leq s \leq 2t$ the extremizers are characterized: each interior shell carries exactly $\lfloor s/2 \rfloor - 1$ balanced-gap minimizing orbits, independently of d and t, with equality only across single such transfers; the two outermost shells contribute a boundary family. A structural theorem then evaluates the sharp layer for the interpolating bodies $C_d + bB^d_\infty$ in every dimension: the sharp constant is determined by lattice-point counts of the same body two dimensions down, together with an explicit facet surcharge for transfers into a zero coordinate, and one inequality in (d, b, s) decides which transfer is minimizing. Equal cubes and equal Lee balls are its two degenerate regimes. Thus convex hyperoctahedral symmetry preserves the transfer sign but neither the sharp scale nor the minimizing orbit; reversals can occur once convexity or coordinate-permutation symmetry fails.

Keywords: lattice cross-covariogram, majorization, hyperoctahedral symmetry, Lee balls, discrete rearrangement, quantitative stability

MSC Classification: 52C07 , 11H06 , 52B20 , 05A20

1 Introduction

For finite lattice sets $A, B \subset \mathbb{Z}^d$, their discrete cross-covariogram is

$$g_{A,B}(u) = |A \cap (B - u)| = |\{x \in A : x + u \in B\}|.$$

Suppose that A and B are cut from compact convex sets invariant under signed coordinate permutations. The value of $g_{A,B}$ then depends only on the decreasing absolute-coordinate composition $|u|^{\downarrow} = (|u|_{(1)}, \ldots, |u|_{(d)})$. Fixing $\|u\|_1$ fixes the translation shell but not this orbit. The basic geometric problem is to compare the overlaps produced by the different compositions on one shell.

The natural comparison is majorization. A unit Robin Hood transfer

$$(a, b) \longmapsto (a - 1, b + 1), \qquad a \geq b + 2, \tag{1}$$

makes two coordinates more balanced while preserving their sum. Three questions arise. First, for which lattice bodies does every such move have a fixed sign? Second, can one recover its exact integer increment and all equality cases rather than only monotonicity? Third, when the increment is strict, which transfer is smallest, what geometric quantity evaluates its sharp value, and how does that quantity move with the shape of the body?

The first question has a strong precedent, which we state at once. Anderson's translation inequality, Mudholkar's group-majorization theorem, and the convolution theorem of Marshall and Olkin give broad continuous Schur orders [1–3]. More pointedly, the decreasing-in-transposition (DT) composition theorem of Hollander, Proschan, and Sethuraman, also applied by them to overlapping sums, allows, in the formulation of Joag-Dev and Sethuraman, any coordinate-permutation-invariant Borel measure for which the composition is well defined [4–6]. Specializing that measure to the lattice Dirac comb already implies the nonnegative sign of an integer balancing transfer, so we do not claim the qualitative order itself as new. What that framework does not produce is an increment, an equality criterion, or an extremal arc, and those are what the second and third questions ask for.

The object of this paper is the exact discrete differential behind the sign. Fix two coordinates and write

$$r = x_i + x_j, \qquad w = x_i - x_j.$$

This rotation of coordinates is standard. Once the residual coordinates and r are fixed, a convex coordinate-permutation-invariant body cuts the w-line in a centered interval, whose lattice points form a centered interval in one coset of $2\mathbb{Z}$. A transfer (1) fixes $R = a + b$ and replaces $D = a - b$ by $D - 2$; on each fiber this shifts one centered parity interval one lattice step toward another, and the overlap change is exactly zero or one. Summing these atomic exposures gives the whole increment (Theorem 3.4). The identity implies the sign in parity-exact form—the cross-covariogram of any two hyperoctahedrally invariant compact convex lattice sections is Schur-concave on every fixed ℓ_1 shell, for possibly different bodies with no rationality assumption—but it also says that an arc is flat if and only if every one of its parity fibers has zero exposure. Everything quantitative below is read off from which fibers are exposed.

The argument needs only the weaker, purely discrete centered exchange-fiber property; convex hyperoctahedral symmetry is a sufficient hypothesis for it.

The quantitative theory is developed first for two equal Lee balls

$$B_1^d(t) = \{x \in \mathbb{Z}^d : \|x\|_1 \le t\},$$

whose translation shells $s = \|u\|_1$ are *active* for $2 \le s \le 2t$. Throughout the paper, a shell is called *interior* when $d \ge 3$ and $4 \le s \le 2t-2$; the remaining active shells are the two outermost ones, $s \in \{2t-1, 2t\}$, and the small shells $s \in \{2,3\}$, and $d=2$ is degenerate in the precise sense that all arcs of a shell then carry the same weight $2t-s+1$. Three theorems describe the weighted majorization graph of an active shell, all for $d \ge 3$. First, an exact evaluation: the least arc weight $\kappa_{d,t,s}$ has a closed form in lower-dimensional Lee-ball counts, obtained from a central-section theorem for integer boxes that is proved by a signed Erdős–Ko–Rado incidence bound, with the even–odd split of the formula reflecting one central box layer versus two adjacent layers. Second, a classification: for $4 \le s \le 2t$ the arcs attaining $\kappa_{d,t,s}$ are determined exactly; on every interior shell they form the balanced-gap axial family, the $\lfloor s/2 \rfloor - 1$ transfers of donor–recipient gap two whose residual mass sits in a single coordinate—a count independent of both $d \ge 3$ and t—while the two outermost shells acquire an explicitly described boundary family. Third, a rigidity theorem: the comparable pairs whose overlap deficit equals $\kappa_{d,t,s}$ times their transfer distance are characterized at every distance, and on interior shells they are single family arcs and nothing else.

The last step is structural. All three sharp models solved here attain their constants inside one set of arcs, the gap-two transfers whose residual mass sits in a single coordinate, indexed by the recipient; what the body decides is which member. For $C_d + bB_\infty^d$ in every dimension $d \ge 3$, on every shell that avoids the outer boundary layer, Theorem 6.4 evaluates the sharp constant by self-overlaps of the *same body two dimensions down*, with an explicit surcharge for transfers into a zero coordinate, and determines the minimizing orbit by the sign of one quantity

$$\Theta_{d,b}(s) = (2b+1)\big(2(2b+1) - s - 1\big) - 2(d-3).$$

Its two terms are a facet contribution and a dimensional one. Only a transfer into a zero coordinate meets the two capped layers of the cube summand, which costs the surcharge; in exchange its residual mass is larger by two, worth an amount proportional to the number of free residual coordinates. When $\Theta_{d,b}(s) > 0$ the minimizer is the member with recipient one, outside the canonical axial-residual orbit that is sharp for Lee balls and cubes; when $\Theta_{d,b}(s) < 0$ it jumps back to the canonical orbit; and both signs, as well as the tie, occur. Equal cubes are the degenerate case of the same computation in which the surcharge is absent, so their unique minimizer is canonical on all active shells; equal Lee balls are the opposite extreme, in which the factorization fails on every arc and the whole balanced-gap family becomes minimizing. One mechanism thus produces three regimes, and the transition between them is located exactly.

Two endpoint phenomena and two boundaries remain: capped cross-polytopes have a tent-shaped crossover in the magnitude of the atomic $(2,0) \to (1,1)$ increment; at fixed axial radius two the three bodies $2B_\infty^3$, $C_3 + B_\infty^3$, and $2C_3$ already have different sharp constants and minimizing orbits (Table 1); sign reversals do occur once convexity or coordinate-permutation symmetry is dropped; and an explicit finite-quotient reversal records a separate arithmetic boundary. No deformation inside the convex hyperoctahedral class can reverse the transfer sign; what moves is the sharp scale and the minimizing orbit. Appendix A lifts the lens order for Lee balls to a cone of radial correlations, where equality is characterized by the support of the mixed-difference coefficients.

1.1 Contributions

The results form the chain summarized in Fig. 1.

1. *Universal parity-exact transfer calculus.* A centered exchange fiber has an explicit endpoint exposure in $\{0, 1\}$. Summing these exposures proves fixed-shell majorization and a fiberwise equality certificate for any pair of hyperoctahedrally invariant compact convex lattice sections.
2. *Sharp Lee stability, classification, and radial equality.* For unequal Lee balls we derive the full two-parity kernel and residual convolution. For equal balls in dimension $d \geq 3$, a central-box theorem evaluates the exact active-shell stability constant by an even–odd formula in lower-dimensional Lee-ball counts, a classification theorem lists all minimizing arcs for $4 \leq s \leq 2t$—on interior shells exactly the $\lfloor s/2 \rfloor - 1$ balanced-gap axial transfers, a count independent of $d \geq 3$ and t—and a rigidity theorem characterizes all pairs attaining equality in the stability inequality, confining them on interior shells to single arcs. The same lens gaps give the order and an exact mixed-difference support criterion for equality in the Lee-radial correlation cone (Appendix A).
3. *A structural theorem with an exact facet threshold.* For the interpolating family $C_d + bB^d_\infty$ in every dimension a master kernel reduces each arc weight to a lower-dimensional lattice-point count plus an explicit facet surcharge carried only by transfers into a zero coordinate. This evaluates the sharp constant and the minimizing orbit on every shell below the boundary layer, and it locates the exact threshold $\Theta_{d,b}(s) = 0$ at which the minimizing orbit changes. Integer cubes and equal Lee balls are the two degenerate regimes of the same mechanism, with, respectively, one and $\lfloor s/2 \rfloor - 1$ minimizing arcs. Capped cross-polytopes add an exact magnitude crossover, and counterexamples exhibit sign reversals once convexity or coordinate-permutation symmetry is removed from the sufficient hypothesis.

1.2 Organization

Section 2 positions the result within covariogram, majorization, and discrete-rearrangement theory. Section 3 proves the universal exchange-fiber calculus. Section 4 develops the sharp Lee specialization and its radial equality theory. Section 5 proves the cube constant and the two interpolation phenomena, and Section 6 the structural

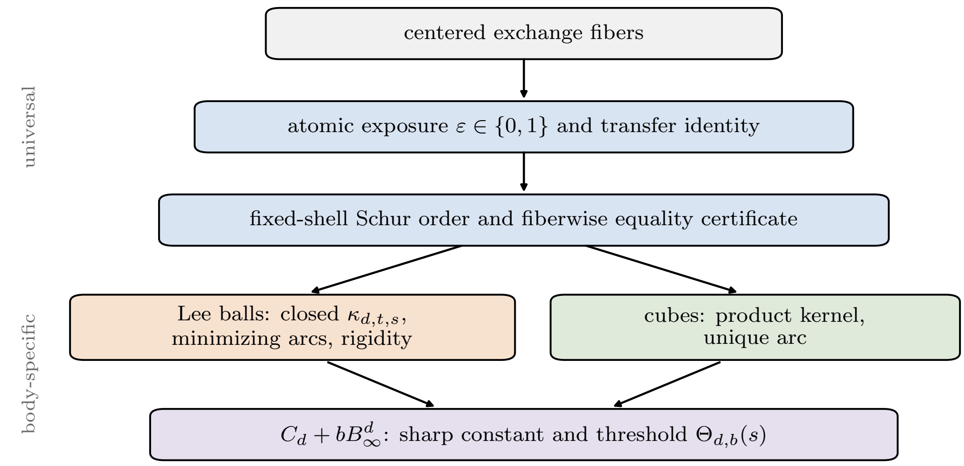

Fig. 1 Logical architecture. Centered exchange fibers produce atomic $\{0,1\}$ exposures, whose sum gives the fixed-shell order and a fiberwise equality certificate. The quantitative layer is body-dependent; the structural theorem of Section 6 evaluates it for $C_d + bB_\infty^d$ and recovers the Lee and the cube regime as the cases without factorization and without surcharge.

evaluation and its facet threshold. Section 7 gives an explicit finite-quotient reversal. The exact lens enumerator and reproduction details are deferred to the appendices.

2 Related work

In each neighbouring literature the overlap function, or a global rearrangement of it, is the object of study; here the two bodies are fixed and known, and the object of study is the order of the overlap values along the majorization graph of one ℓ_1 shell, resolved edge by edge into exact integer increments.

2.1 Covariograms, cross-covariograms, and translated intersections

The function

$$u \mapsto J_{p,q}^{(d)}(u) = \left|B_1^d(p) \cap (B_1^d(q) - u)\right|$$

is a discrete cross-covariogram; more generally this paper studies $u \mapsto |(K \cap \mathbb{Z}^d) \cap ((L \cap \mathbb{Z}^d) - u)|$ for two known symmetric convex bodies K, L. The continuous covariogram originated in random-set and integral geometry, and geometric tomography made the inverse question—which bodies are determined by all translated-overlap volumes?—a central theme [7–9]. For planar convex bodies that program developed through partial results and the resolution of Matheron's conjecture [10, 11], with a complementary phase retrieval formulation [12], and unequal sets have their own reconstruction problem [13]. Translated intersections also appear in direct geometric inequalities: the radial dependence of the continuous intersection volume is tied to ellipsoids [14], and recent unpublished work of Bianchi, Burchard, and Lin (arXiv:2508.03887) studies strict concavity of cross-covariograms along Euclidean segments.

For finite lattice sets, discrete covariograms were developed in connection with sums, projections, sections, and reconstruction [15, 16]; homometric lattice-convex sets show that the complete discrete covariogram can have nontrivial equality classes [17, 18]. Alonso-Gutiérrez, Lucas Marín, and Martín Goñi studied the lattice-point-enumerator covariogram $\widetilde{g}_K(x) = \mathrm{G}_n\big(K \cap (x + K)\big)$ and the associated covariogram ball bodies in connection with discrete projection inequalities of Zhang type [19]; their results concern radial integrals and inclusions between ball bodies. Discrete tomography reconstructs finite sets from X-rays, with a corresponding notion of stability under noisy or inconsistent projection data [20, 21]. Our equality classes are not homometric pairs and our stability is not robustness to perturbed data; both are intrinsic to the values of one fixed cross-covariogram along the integer unit-transfer graph.

2.2 Lee geometry, compositions, and lattice tilings

Lee introduced the nonbinary metric [22] and its channel-theoretic form followed soon afterwards [23]. In finite Lee spaces the natural association relations are indexed by Lee compositions rather than by scalar Lee distance alone, a distinction underlying the Lee scheme and its linear-programming theory [24, 25]: composition sensitivity is itself classical. Our refinement fixes the total Lee length of a translation on $\mathbb{Z}^d$ and resolves the change between comparable compositions into exact increments. Tiling and packing by Lee balls [26–31] and perfect, quasi-perfect, diameter-perfect, and dense Lee codes [32–35] constrain disjoint or near-disjoint families of translates rather than comparing the lens deficits of two translations on one shell. Explicit finite-q intersection formulas do occur in identifying-code bounds [36] and through generating functions for Lee-ball volumes [37], but not the fixed-shell comparison structure studied here.

One recent preprint is a close methodological neighbour. Albader (arXiv:2606.17527) studies local fault repair of perfect resource placements in dense two-dimensional Gaussian networks and, for that repair geometry, proves exact replacement numbers together with the sharp minimum-overlap formula $\Omega_G(t) = t + 1$ for $t \geq 2$ among minimum-size repairs. The overlap lower bound there is obtained in the rotated coordinates $u = x + y$, $v = x - y$, in which planar Lee balls become parity-constrained squares: exactly the two-dimensional case of the sum–difference slicing used throughout Section 3 and of the diamond-to-square parity treatment opening Section 4, and both arguments exploit two parity classes to obtain exact counts. The results are nevertheless disjoint. The optimization there is over replacement placements repairing a failed Lee cell in the plane, and its extremal quantity is a minimum overlap among minimum-size repairs; here the two bodies are fixed, the variable is their relative integer translation on a prescribed ℓ_1 shell in every dimension $d \geq 3$, and the extremal data are the sharp transfer constant $\kappa_{d,t,s}$, the minimizing arcs, and the equality pairs of a stability inequality. The rotation is a shared tool, not a shared theorem.

2.3 Rearrangement, compression, and quantitative stability

Anderson proved monotonicity of the integral of a symmetric unimodal function over a translated symmetric convex set [1]. Mudholkar replaced central symmetry by invariance under a finite measure-preserving linear group and obtained group-majorization, including permutation and signed-permutation cases [2]; Marshall and Olkin proved convolution preservation of Schur-concavity for continuous multivariate densities [3]. These continuous results rule out a volume-level sign transition between the cross-polytope and cube.

The closest qualitative antecedent on the lattice comes from DT functions. Hollander, Proschan, and Sethuraman established their composition properties [4] and in a companion paper proved the DT property of *overlapping sums* of independent random variables [5]. Theorems 4–5 of Joag-Dev and Sethuraman formulate the composition theorem for any coordinate-permutation-invariant Borel measure for which the composition is well defined [6]; taking that measure to be $\sum_{z\in\mathbb{Z}^d}\delta_z$ and the two kernels to be indicators of symmetric convex bodies yields the nonnegative sign of every lattice balancing transfer. As announced in the introduction, we do not claim that sign, that is, the inequality of Corollary 3.5, as new. What the preservation route does not produce is an increment value, an equality criterion, or an extremal set: it returns one inequality per comparison, whereas Theorem 3.4 returns the identity (6) expressing that comparison as a sum of $\{0,1\}$-valued parity-fiber exposures over explicitly indexed lattice fibers, and forgetting which fibers are exposed recovers the DT conclusion. To the best of our knowledge that identity does not appear in the DT literature. The results below all depend on which fibers are exposed: the fiberwise equality certificate of Corollary 3.5, the closed constants of Theorems 4.17 and 5.1, the extremal classifications of Theorems 4.25 and 4.26, and the structural evaluation of Section 6.

The centered exchange-fiber property should also be compared with existing discrete set classes. It is a sectional condition: every two-coordinate fiber, in sum–difference coordinates, must be a centered parity interval. It is implied by convex hyperoctahedral lattice sections, but it is not an exchange axiom in the sense of discrete convex analysis. The M-convex and jump-system exchange axioms [38, 39] constrain how one point of a set can move toward another point of the *same* set and impose no symmetry, whereas the present property constrains the shape and centering of one-dimensional parity sections and imposes no exchange between distinct points. Neither condition subsumes the other: the set $\{(0,0),(2,2)\}\subset\mathbb{Z}^2$ has centered exchange fibers (both difference fibers are the single point $\{0\}$) but violates the two-step exchange axiom of a jump system, since no lattice point adjacent to $(0,0)$ toward $(2,2)$ admits a second step into the set, and for the same reason it is not M$^\natural$-convex; conversely, the singleton $\{(1,0)\}$ is both a trivial M-convex set and a trivial jump system, yet its difference fiber $\{1\}$ is not centered. We therefore use a separate name.

Classical symmetric-decreasing rearrangement compares a function or a set with a centered equimeasurable profile; the Riesz inequality and its multilinear extensions are standard instances [40–42]. The abstract vector order used here is classical majorization, equivalently generated by elementary balancing transforms [43, 44]. Equality in continuous Riesz rearrangement and local symmetrization by polarization have their own theories [45, 46], and monotone or supermodular integrands provide a broader functional setting [47]. Discrete convolution and Schwarz rearrangements have been developed on graphs and integer lattices [48–50], and coordinate compression is a classical local-to-global method in discrete isoperimetry [51, 52]. All of these move the sets or functions: a rearrangement recenters them, and compression changes a set to improve a boundary or shadow. Recentering in particular erases the distinction between the translations $(2,0)$ and $(1,1)$, which is precisely the fixed-shell variable retained here. Our local move acts on the integer translation partition rather than on either radial profile, and its effect is an exact finite difference.

Quantitative stability normally asks whether an analytic deficit controls distance from an extremal orbit. Sharp quantitative isoperimetry and its transport developments and surveys provide prototypes [53–55]; stability for Riesz rearrangement gives a closer continuous correlation analogue [56]. Discrete lattice stability results typically describe large near-minimizers by their proximity to structured sets such as zonotopes [57]. Our statements have a different exact finite structure: on one fixed integer shell the metric is the minimum number of unit Robin Hood transfers, the deficit is a sum of explicit integer arc weights, and the extremal constants, arcs, and pairs are evaluated and classified rather than bounded.

2.4 Lattice-point enumeration and the finite-quotient boundary

The normalization $B_1^d(t) = tC_d \cap \mathbb{Z}^d$ places Lee balls within the standard theory of integer-point enumeration in polytopes and generating functions [58, 59], and cross-polytopes have been studied through their lattice points, intrinsic volumes, and packing density [60, 61]. Discrete Brunn–Minkowski inequalities and lattice-point enumerator bounds compare counts under sums, dilations, sections, or projections [62–64], whereas our sets and radii are fixed. One structure from that circle of ideas does reappear as a conclusion rather than a hypothesis: the sharp constants of Section 6 are computed from lattice-point counts of the same body in dimension $d-2$. Finally, the universal theorem below uses the centered interval cut out by a fixed coordinate-sum line together with the parity sublattice of $\mathbb{Z}^d$; wrap-around destroys that slicing, and Section 7 shows that the naive finite-q extension is false.

3 Coordinate balancing for symmetric lattice bodies

Throughout the paper the notation is divided as follows: K, L denote compact convex sets in $\mathbb{R}^d$ and A, B finite lattice sets, typically their lattice sections; u, v are integer shift vectors and λ, μ their decreasing absolute-coordinate compositions on one shell; a, b are the two entries changed by a unit transfer (1) and w is the residual shift vector of the untouched coordinates; p, q, t are radii, s is a shell weight, and K inside a binomial coefficient is a truncation level, never a body.

We first isolate the part of the argument that does not depend on Lee balls. For a finite set $A \subset \mathbb{Z}^d$, coordinates $i < j$, a residual vector $z \in \mathbb{Z}^{d-2}$, and $r \in \mathbb{Z}$, define the difference fiber

$$\mathcal{I}_A^{ij}(z, r) = \left\{ \delta \in r + 2\mathbb{Z} : x_{-\{i,j\}} = z,\ x_i = \frac{r+\delta}{2},\ x_j = \frac{r-\delta}{2},\ x \in A \right\}. \tag{2}$$

The parity restriction is the image of $\mathbb{Z}^2$ under $(x_i, x_j) \mapsto (x_i + x_j, x_i - x_j)$.

Definition 3.1 (Centered exchange fibers) We say that A has the *centered exchange-fiber property* if every nonempty set in (2) is of the form

$$\{-m, -m+2, \ldots, m-2, m\} \qquad (m \in \mathbb{Z}_{\geq 0},\ m \equiv r \pmod 2). \tag{3}$$

No convex hull is mentioned in this definition; it is a finite, directly checkable property of a lattice set. Figure 2 shows a planar instance.

For two finite sets put

$$g_{A,B}(u) = |A \cap (B-u)| = |\{x \in A : x+u \in B\}|.$$

Proposition 3.2 (Convex symmetric bodies have centered exchange fibers) *Let $K \subset \mathbb{R}^d$ be a compact convex set invariant under coordinate permutations. Then $K \cap \mathbb{Z}^d$ has the centered exchange-fiber property. If K is invariant under signed coordinate permutations, then $K \cap \mathbb{Z}^d$ is invariant under the same group. Moreover, if K and L are both invariant under that group, then*

$$g_{K\cap\mathbb{Z}^d, L\cap\mathbb{Z}^d}(\gamma u) = g_{K\cap\mathbb{Z}^d, L\cap\mathbb{Z}^d}(u)$$

for every signed coordinate permutation γ and every $u \in \mathbb{Z}^d$.

Proof Fix i, j, z, r. Intersecting K with the affine line on which the residual coordinates and $x_i + x_j = r$ are fixed gives an interval, possibly empty or a singleton. The transposition $i \leftrightarrow j$ preserves this line and maps $\delta = x_i - x_j$ to $-\delta$, so the interval is centered. Its lattice points have $\delta \equiv r \pmod 2$ and therefore form exactly the parity interval (3). The last assertion follows by applying the same signed permutation to the summation variable and to both bodies. $\square$

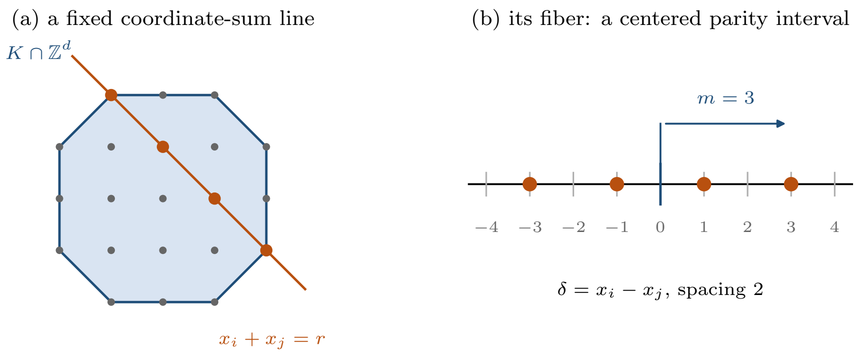

Fig. 2 The exchange fiber of Definition 3.1 in a symmetric planar section. Fixing the residual coordinates and the sum $x_i + x_j = r$ cuts out a segment of the body; its lattice points, recorded through $\delta = x_i - x_j$, form a centered interval of one parity class. Here the four marked points have $\delta \in \{-3, -1, 1, 3\}$, so $m = 3$.

The atomic fact behind the general theorem is one-dimensional and exact; the content lies in the fiberwise assembly and its quantitative consequences. Figure 3 shows its three regimes.

Lemma 3.3 (Atomic parity-interval exposure) *Let*

$$I_\alpha = \{-\alpha, -\alpha+2, \dots, \alpha\}, \qquad I_\beta = \{-\beta, -\beta+2, \dots, \beta\},$$

where $\alpha, \beta \in \mathbb{Z}_{\geq 0}$*, and assume that* I_α *and* $I_\beta - D$ *lie in the same coset of* $2\mathbb{Z}$*. For every integer* $D \geq 2$*,*

$$\varepsilon_{\alpha,\beta}(D) := |I_\alpha \cap (I_\beta - (D-2))| - |I_\alpha \cap (I_\beta - D)| \tag{4}$$

$$= \mathbf{1}\{|\beta - D + 2| \leq \alpha < \beta + D\} \in \{0, 1\}. \tag{5}$$

If either interval is empty, define its exposure to be zero.

Proof The shifted parity intervals differ only at their two endpoints:

$$I_\beta - (D-2) = \big((I_\beta - D) \setminus \{-\beta - D\}\big) \cup \{\beta - D + 2\}.$$

Consequently the overlap change equals

$$\mathbf{1}\{|\beta - D + 2| \leq \alpha\} - \mathbf{1}\{\beta + D \leq \alpha\}.$$

Because $|\beta - D + 2| \leq \beta + D$ for $\beta \geq 0$ and $D \geq 2$, the second indicator can be one only when the first is one. Their difference is the indicator in (5). The parity-compatibility hypothesis ensures that every endpoint displayed is in the common parity coset. $\square$

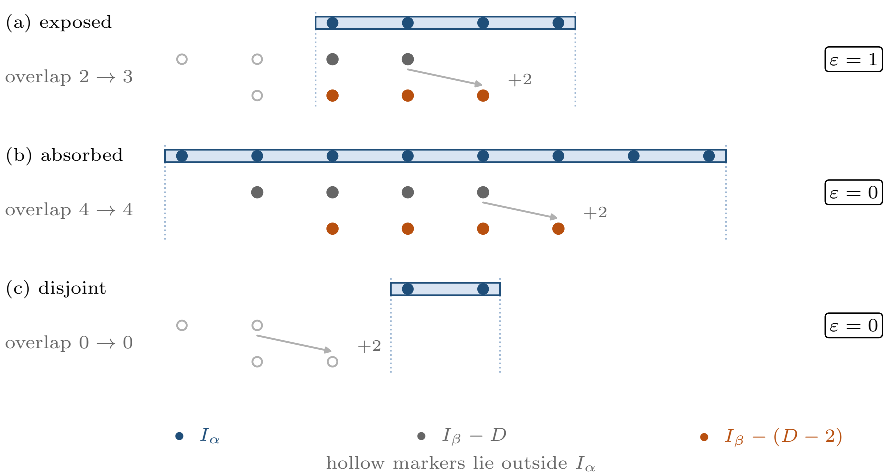

Fig. 3 The three regimes of Lemma 3.3. Moving the parity interval $I_\beta - D$ two steps toward I_α removes its left endpoint and adds a new right endpoint, so the overlap can grow by at most one point. It grows by exactly one when the new endpoint enters I_α while the old one was already outside (a); it is unchanged when I_α contains both (b) or neither (c).

Summing this exposure over the fibers gives the transfer identity.

Theorem 3.4 (Exact exchange-fiber transfer) *Let* $A, B \subset \mathbb{Z}^d$ *be finite sets with centered exchange fibers. Fix coordinates* $i < j$ *and a shift* $u \in \mathbb{Z}^d$*, written* $u = (a, b, u_{-\{i,j\}})$ *in* *coordinate-labelled form, whose two active entries $a = u_i$ and $b = u_j$ are integers with $a \geq b+2$; the residual entries $u_{-\{i,j\}} \in \mathbb{Z}^{d-2}$ are arbitrary. Put*

$$u' = u - e_i + e_j, \qquad R = a + b, \qquad D = a - b.$$

For a nonempty fiber write $m_A^{ij}(z, r)$ for its nonnegative endpoint, and use the analogous notation for B. If a fiber is empty, put its endpoint equal to the symbol $\varnothing$ and set $\varepsilon_{\varnothing,\beta} = \varepsilon_{\alpha,\varnothing} = 0$. Then

$$g_{A,B}(u') - g_{A,B}(u) = \sum_{z\in\mathbb{Z}^{d-2}} \sum_{r\in\mathbb{Z}} \varepsilon_{m_A^{ij}(z,r),\, m_B^{ij}(z+u_{-\{i,j\}},r+R)}(D), \tag{6}$$

In particular, every summand belongs to $\{0, 1\}$ and the increment is nonnegative. The transfer is flat if and only if every fiber fails the endpoint-exposure condition (5).

Proof Fix z and write $r = x_i + x_j$, $\delta = x_i - x_j$. The first membership condition is $\delta \in \mathcal{I}_A^{ij}(z, r)$. For $x + u \in B$, the translated coordinate sum and difference are $r + R$ and $\delta + D$, so the second condition is

$$\delta + D \in \mathcal{I}_B^{ij}(z + u_{-\{i,j\}}, r + R).$$

Thus this slice contributes the overlap of two centered parity intervals, the second translated by $-D$. Their parity cosets agree because $R - D = 2b$. The transfer fixes R and replaces D by $D - 2$. Lemma 3.3 gives the displayed exposure for this slice; summing the disjoint slices proves (6). Since all summands are nonnegative, their sum vanishes exactly when each one does. □

For nonnegative integer vectors of equal coordinate sum, write $x \succeq y$ when

$$\sum_{k=1}^{h} x_k^{\downarrow} \geq \sum_{k=1}^{h} y_k^{\downarrow} \qquad (1 \leq h < d).$$

Corollary 3.5 (Symmetric lattice cross-covariogram majorization) *Let $K, L \subset \mathbb{R}^d$ be compact convex sets invariant under signed coordinate permutations, and put $A = K \cap \mathbb{Z}^d$, $B = L \cap \mathbb{Z}^d$. If integer shifts $u, v \in \mathbb{Z}^d$ satisfy*

$$|u|^{\downarrow} \succeq |v|^{\downarrow}, \qquad \|u\|_1 = \|v\|_1, \tag{7}$$

then

$$g_{A,B}(u) \leq g_{A,B}(v). \tag{8}$$

The same conclusion holds for any signed-permutation-invariant finite sets A, B with centered exchange fibers. For comparable u, v, equality holds if and only if all exposures vanish on one—equivalently, on every—monotone unit-transfer chain from $|u|^{\downarrow}$ to $|v|^{\downarrow}$.

Proof Signed-permutation invariance reduces to nonnegative decreasing shifts; Theorem 3.4 itself imposes no sign condition on the entries. Integer majorization is generated by unit Robin Hood transfers followed by permutations [44]; see also Lemma 4.7 for a self-contained construction. Apply Theorem 3.4 along such a chain. The endpoint difference is a sum of nonnegative arc weights, so it is zero exactly when every arc on the chain has zero weight. The reason one chain suffices is that the endpoint difference $g_{A,B}(v) - g_{A,B}(u)$ does not depend on the chain: if the arc weights of one monotone chain sum to zero, the same number is the sum of the nonnegative arc weights of any other monotone chain, whose arcs must therefore all be flat as well. □

Throughout the paper we distinguish three levels of equality language. *Local edge equality* means that one unit-transfer arc has weight zero; Theorem 3.4 characterizes it fiberwise. *Comparable endpoint equality* means $g_{A,B}(u) = g_{A,B}(v)$ for a comparable pair (7); by Corollary 3.5 it is equivalent to local edge equality along one—equivalently every—monotone chain. *Extremizer uniqueness* asks whether the minimizing composition of a shell is unique modulo signed permutations. All three levels are relative to the majorization order: strict Schur-concavity is an order-theoretic statement about comparable pairs, and incomparable shell compositions may take equal overlap values (Remark 4.10). For a general body, the fiberwise conditions above are a *certificate*: a prescribed transfer or chain can be tested fiber by fiber, but no geometric description of the flat transfers is asserted. The *geometric classifications* of extremal arcs and pairs proved in this paper are body-specific results: Theorems 4.25 and 4.26 for Lee balls and Corollary 5.2 for cubes.

Remark 3.6 (Relation to DT composition) The weak nonnegative sign in Corollary 3.5 can alternatively be recovered from the classical DT preservation framework of Hollander, Proschan, and Sethuraman, in the formulation of Joag-Dev and Sethuraman [4–6]. We do not use that route in any proof below: Theorem 3.4 resolves the increment into integer endpoint exposures and thereby supplies the strictness, equality, and sharp-stability information that the preservation framework does not record.

The set theorem also has a functional form. It is useful when the profiles are not indicators of one body.

Corollary 3.7 (Exchange-fiber profile correlations) *Let $f, g : \mathbb{Z}^d \to \mathbb{R}_{\geq 0}$ be finitely supported and invariant under signed coordinate permutations. Suppose every positive superlevel set of f and of g has centered exchange fibers. Then*

$$u \longmapsto \sum_{x \in \mathbb{Z}^d} f(x) g(x+u)$$

is Schur-concave in $|u|^{\downarrow}$ on each fixed ℓ_1 shell. In particular, this applies when the superlevel sets are lattice sections of hyperoctahedrally invariant compact convex sets.

Proof Use the layer-cake identities $f(x) = \int_0^\infty \mathbf{1}_{\{f(x)>s\}}\, ds$ and similarly for g. Tonelli's theorem expresses the correlation as a nonnegative double integral of cross-covariograms of superlevel sets. Apply Corollary 3.5 pointwise in the two levels. □

The centered exchange-fiber condition is sufficient, not asserted to be necessary. The following examples show that neither ingredient in the convex hyperoctahedral hypothesis can be dropped wholesale.

Example 3.8 (Sharpness examples for the hypotheses) In dimension two, let

$$X = \{0,\ \pm e_1,\ \pm e_2,\ \pm 2e_1,\ \pm 2e_2\}.$$

This finite set is unconditional and permutation invariant, but its diagonal fiber at $r = 2$ has a hole. Direct counting gives

$$g_{X,X}(2,0) = 3 > 2 = g_{X,X}(1,1),$$

so balancing has increment -1. Filling the diagonal holes to obtain $B_1^2(2)$ changes the same increment to $8 - 5 = 3$.

Conversely, convexity without coordinate-permutation symmetry is insufficient. For the lattice rectangle $Y = ([-5,5] \times [-1,1]) \cap \mathbb{Z}^2$,

$$g_{Y,Y}(2,0) = 9 \cdot 3 = 27 > 20 = 10 \cdot 2 = g_{Y,Y}(1,1).$$

These counterexamples show that a sign reversal becomes possible outside the sufficient class above; they do not claim that every set outside that class must reverse.

On any finite shell on which all unit-transfer arcs are strict, their least integer weight is automatically a sharp slope against shortest unit-transfer distance for comparable pairs. Determining that least weight is not universal: it is the body-specific third layer developed next.

4 Lee-ball calculus and sharp stability

For $d \geq 1$ and $t \in \mathbb{Z}_{\geq 0}$, let

$$C_d = \{x \in \mathbb{R}^d : \|x\|_1 \leq 1\}, \qquad B_1^d(t) = tC_d \cap \mathbb{Z}^d.$$

Its lattice count is

$$L_d(t) = |B_1^d(t)| = \sum_{j=0}^{\min(d,t)} 2^j \binom{d}{j}\binom{t}{j}. \tag{9}$$

An exact all-scale rational enumerator for the equal-radius lens—a closed bivariate generating function together with a binomial extraction formula—is stated and proved as Theorem B.1 in Appendix B. It is not needed for the balancing calculus and reappears only once, as an alternative derivation of one atomic identity in the proof of Theorem 4.17. One of its special cases is worth recording here:

$$|B_1^2(2) \cap (B_1^2(2) - 2e_1)| = 5 \qquad \text{whereas} \qquad |B_1^2(2) \cap (B_1^2(2) - e_1 - e_2)| = 8,$$

so two shifts of the same Lee weight can have different lattice lenses, and scalar Lee distance does not determine the overlap.

4.1 Exact balancing calculus

Recall that for $\lambda, \mu \in \mathbb{Z}_{\geq 0}^d$ with equal coordinate sum, we write $\lambda \succeq \mu$ when their decreasing rearrangements satisfy

$$\sum_{i=1}^{k} \lambda_i^{\downarrow} \geq \sum_{i=1}^{k} \mu_i^{\downarrow} \qquad (1 \leq k < d).$$

Thus $\lambda \succeq \mu$ means that λ is at least as coordinate-concentrated as μ. For radii $p, q \in \mathbb{Z}_{\geq 0}$, define the unequal-radius Lee lens

$$J_{p,q}^{(d)}(u) = \left|\{x \in \mathbb{Z}^d : \|x\|_1 \leq p,\ \|x+u\|_1 \leq q\}\right|.$$

Figure 4 shows the elementary move underlying the order: one unit passes from the larger coordinate to the smaller coordinate while their sum remains fixed.

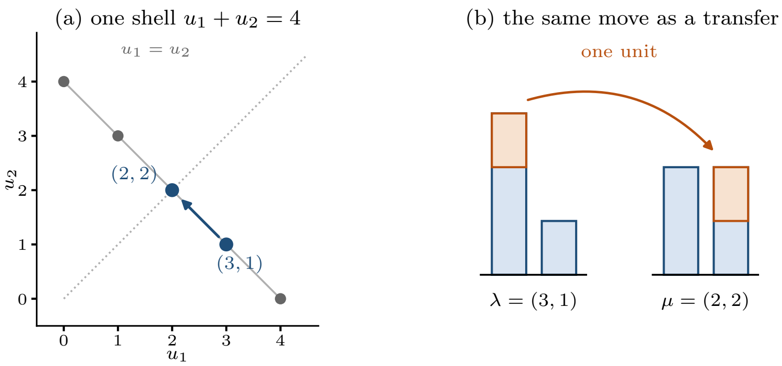

Fig. 4 A unit balancing transfer in translation-composition space. The shifts $u = (3, 1)$ and $u' = (2, 2)$ lie on the same shell $u_1 + u_2 = 4$, and the move sends one unit from the larger coordinate to the smaller one.

For $p, q \in \mathbb{Z}_{\geq 0}$, $c \in \mathbb{Z}$, and $\epsilon \in \{0, 1\}$, let

$$n_\epsilon(p, q; c) = |[-p, p] \cap [-c - q, -c + q] \cap (2\mathbb{Z} + \epsilon)|. \tag{10}$$

Extend $n_\epsilon(p, q; c)$ by zero whenever $p < 0$ or $q < 0$. For nonnegative radii these are elementary interval counts. If $\ell = \max\{-p, -c - q\}$ and $h = \min\{p, -c + q\}$, let $\rho_\epsilon(\ell) \in \{0, 1\}$ be the residue satisfying $\rho_\epsilon(\ell) \equiv \epsilon - \ell \pmod 2$. The first integer of parity ϵ at or above ℓ is $\ell + \rho_\epsilon(\ell)$, and hence

$$n_\epsilon(p, q; c) = \max\left\{0, 1 + \left\lfloor \frac{h - \ell - \rho_\epsilon(\ell)}{2} \right\rfloor\right\}. \tag{11}$$

Thus the parameter space is cut by finitely many affine endpoint and parity walls. The next theorem evaluates the increment of a balancing step.

Theorem 4.1 (Exact two-coordinate balancing calculus) *For $p, q, a, b \in \mathbb{Z}_{\geq 0}$ with $a \geq b + 2$, put*

$$M_{p,q}(a, b) = \left|\left\{(x, y) \in \mathbb{Z}^2 : |x| + |y| \leq p,\ |x + a| + |y + b| \leq q\right\}\right|.$$

Write $R = a + b$ and $D = a - b$. Then

$$M_{p,q}(a - 1, b + 1) - M_{p,q}(a, b) = \Phi_{p,q}(R, D) \tag{12}$$

$$:= \sum_{\epsilon=0}^{1} n_\epsilon(p,q;R)\big(n_\epsilon(p,q;D-2) - n_\epsilon(p,q;D)\big). \tag{13}$$

Each bracket in (13) belongs to $\{0,1\}$. In particular $\Phi_{p,q}(R,D) \geq 0$, and equality holds if and only if

$$n_\epsilon(p,q;R)\big(n_\epsilon(p,q;D-2) - n_\epsilon(p,q;D)\big) = 0 \quad (\epsilon = 0,1). \tag{14}$$

For radii that differ by at most one, the chamber formula collapses to

$$\Phi_{p,q}(R,D) = \begin{cases} \frac{1}{2}(p+q-R+1)_+, & |p-q| = 1 \text{ and } D = 2, \\ (p+q-R+1)_+, & \text{otherwise}, \end{cases} \qquad (|p-q| \leq 1). \tag{15}$$

In particular, $\Phi_{t,t}(R,D) = (2t-R+1)_+$, so every equal-radius balancing move is strict exactly when $a+b \leq 2t$.

Proof Use the parity-preserving diamond-to-square coordinates

$$r = x+y, \qquad w = x-y.$$

They biject $\mathbb{Z}^2$ with $\Lambda = \{(r,w) \in \mathbb{Z}^2 : r \equiv w \pmod 2\}$, and

$$|x| + |y| = \max\{|r|, |w|\}.$$

The radius-two example in Fig. 5 displays the bijection and the two interlaced parity classes directly.

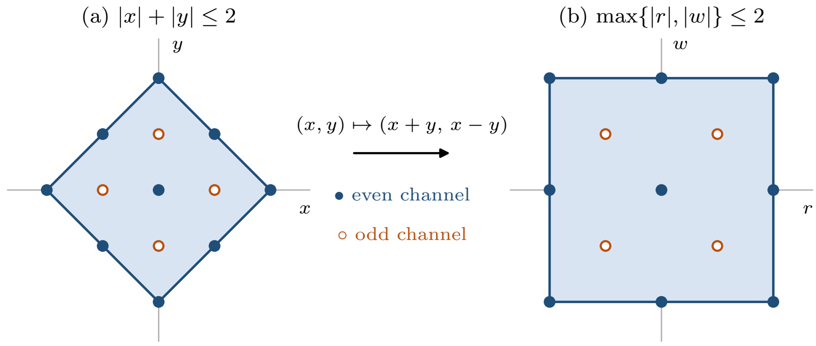

Fig. 5 The diamond-to-square map. The transformation $T(x,y) = (x+y, x-y)$ sends the integer points of $B_1^2(2)$ to the points of $[-2,2]^2$ lying in $\Lambda = \{(r,w) \in \mathbb{Z}^2 : r \equiv w \pmod 2\}$. Filled and open circles distinguish the even–even and odd–odd channels, which the map keeps separate.

Writing $R = a+b$ and $D = a-b$, the second norm is

$$\max\{|r+R|, |w+D|\}.$$

Consequently the points being counted are

$$\big(I(R) \times I(D)\big) \cap \Lambda, \qquad I(c) = [-p,p] \cap [-c-q, -c+q] \cap \mathbb{Z}. \tag{16}$$

The balancing step fixes R and replaces $D \geq 2$ by $D-2$. Splitting Λ into its even–even and odd–odd classes gives

$$M_{p,q}(a,b) = \sum_{\epsilon=0}^{1} n_\epsilon(p,q;R)\, n_\epsilon(p,q;D),$$

and subtraction proves the parity expansion (13), hence (12).

It remains to identify the sign without imposing any relation between p and q, and this is exactly the atomic exposure of Lemma 3.3. Fix a parity class ϵ. The section $[-p,p]\cap(2\mathbb{Z}+\epsilon)$ is a centered parity interval: it equals I_α with

$$\alpha = \max\{n\in\mathbb{Z}_{\geq 0} : n\leq p,\ n\equiv \epsilon \pmod 2\},$$

and it is empty exactly when no such n exists. Likewise, $z\in[-c-q,-c+q]\cap(2\mathbb{Z}+\epsilon)$ if and only if $z+c\in[-q,q]\cap(2\mathbb{Z}+\epsilon+c)$, so with

$$\beta = \max\{n\in\mathbb{Z}_{\geq 0} : n\leq q,\ n\equiv \epsilon+c \pmod 2\}$$

one has $[-c-q,-c+q]\cap(2\mathbb{Z}+\epsilon)=I_\beta-c$, again with the empty convention. Hence

$$n_\epsilon(p,q;c) = |I_\alpha\cap(I_\beta-c)|.$$

Figure 6 shows a chamber in which one parity channel gains a point while the other exchanges one point without changing its count.

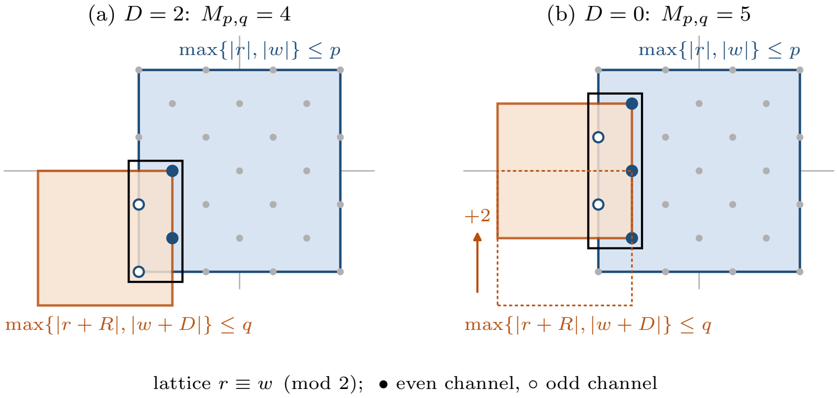

Fig. 6 The balancing step in square coordinates for $p=3$, $q=2$, and $(3,1)\mapsto(2,2)$, so $R=4$ and D drops from 2 to 0. Both bodies become axis-parallel squares, the counted set is the rectangle $I(R)\times I(D)$ intersected with Λ, and the move translates the second square by two in w. Here $I(4)=\{-3,-2\}$ is fixed while the moving interval changes from $I(2)=\{-3,-2,-1,0\}$ to $I(0)=\{-2,-1,0,1,2\}$; the channel counts pass from $(2,2)$ to $(3,2)$, so the odd channel loses one point and gains another and the picture asserts no set containment.

Now specialize to $c=D$ and $c=D-2$. Since $D\equiv D-2 \pmod 2$, the same β serves both displacements, and $I_\beta-D$ lies in the coset $2\mathbb{Z}+\epsilon$ of I_α. Lemma 3.3, together with its zero convention for empty intervals, therefore gives directly

$$n_\epsilon(p,q;D-2)-n_\epsilon(p,q;D) = \varepsilon_{\alpha,\beta}(D)\in\{0,1\}.$$

Since both factors in (13) are nonnegative, this also proves (14).

If $|p-q|\leq 1$ and $c\geq 1$, then $I(c)=[-p,q-c]\cap\mathbb{Z}$ whenever this interval is nonempty. Except when $|p-q|=1$ and $D=2$, passing from D to $D-2$ adds one point in each parity class, so the total contribution is $|I(R)|=(p+q-R+1)_+$. In the exceptional case R is even. Whenever $I(R)$ is nonempty, its length $p+q-R+1$ is even, it contains equally many points of each parity, and a direct comparison of $I(2)$ with $I(0)$ shows that exactly one parity bracket is one. The contribution is therefore half that length. If $R>p+q$, then $I(R)$ is empty in every case. This proves (15) and its equal-radius specialization. □

In every subsequent residual formula, we extend $\Phi_{p,q}(R,D)$ by zero whenever $p<0$ or $q<0$.

4.1.1 Boundary gaps

Theorem 4.1 intentionally assumes $D = a - b \geq 2$, which is exactly the domain of a nontrivial integer Robin Hood transfer. If $D = 0$, there is no ordered donor coordinate. If $D = 1$, the putative move maps (a, b) to (b, a) and therefore remains in the same permutation orbit, so its lens gap is zero. Neither case is an edge of the strict majorization graph. Empty chambers require no extra convention, including those that can arise at boundary choices such as $p = 0$ or $q = 0$: whenever the clipped interval is empty, the maximum with zero in (11) makes the corresponding parity count vanish. Thus the theorem covers all unequal radii on its full transfer domain. The same uniform conventions seal every small-parameter case appearing later: for $d = 2$ the closed formulas never invoke L_{d-3}, because the two-dimensional line of (29) is stated separately; for $d = 3$ and $s = 3$ the convention $L_0(r) = 1$ applies; for $t = 1$ the only active Lee shell is $s = 2$; the cube formulas of Theorem 5.1 extend through $s = 2t + 1$ with vanishing factors allowed; the mixed family of Theorem 5.4 includes $b = 1$ with its one extra minimizing arc; and empty parity sections always fall under the empty-interval convention of Lemma 3.3. No boundary case is omitted. Figures 4– 6 separate the three levels of the mechanism: shell motion, the square-coordinate bijection, and the parity-resolved chamber.

The two-dimensional law lifts without loss. For an untouched shift $w \in \mathbb{Z}^{d-2}$ define its residual histogram

$$h_w(r, s) = |\{y \in \mathbb{Z}^{d-2} : \|y\|_1 = r,\ \|y + w\|_1 = s\}|. \tag{17}$$

Proposition 4.2 (Residual-convolution differential law) *Suppose $p, q \in \mathbb{Z}_{\geq 0}$ and $u = (a, b, w) \in \mathbb{Z}^2_{\geq 0} \times \mathbb{Z}^{d-2}$ with $a \geq b + 2$, and let $u' = (a - 1, b + 1, w)$. Then*

$$J^{(d)}_{p,q}(u') - J^{(d)}_{p,q}(u) = \sum_{r,s \geq 0} h_w(r, s)\Phi_{p-r,q-s}(a + b, a - b), \tag{18}$$

where the zero-extension convention for a negative residual radius applies. Hence the increment is nonnegative. It is zero if and only if every summand with $h_w(r, s) > 0$ has both parity channels in (14) equal to zero.

Proof Fix the untouched vector y. Its two norms consume $r = \|y\|_1$ and $s = \|y + w\|_1$, leaving the two-coordinate radii $p - r$ and $q - s$. Theorem 4.1 gives the increment on that slice. Grouping slices with the same (r, s) proves (18). Every term is nonnegative, so the asserted equality criterion is exact. □

Lemma 4.3 (Balanced residual slice) *Let $t \in \mathbb{Z}_{\geq 1}$, let $u = (a, b, w) \in \mathbb{Z}^d$ with $a \geq b + 2 \geq 2$, and put $u' = (a - 1, b + 1, w)$. If*

$$a + b + \|w\|_1 \leq 2t,$$

then there is a residual slice $y \in \mathbb{Z}^{d-2}$ whose two remaining radii

$$p' = t - \|y\|_1, \qquad q' = t - \|y + w\|_1$$

are nonnegative, satisfy $|p' - q'| \leq 1$ and $p' + q' \geq a + b$, and have

$$\Phi_{p',q'}(a + b, a - b) > 0.$$

Consequently $J^{(d)}_{t,t}(u') > J^{(d)}_{t,t}(u)$.

Proof Set $c = \|w\|_1$. Choose integers $0 \le r_k \le |w_k|$ with $\sum_k r_k = \lfloor c/2 \rfloor$, which is possible because the capacities $|w_k|$ sum to c, and put $y_k = -\operatorname{sgn}(w_k) r_k$. Then

$$\|y\|_1 = \lfloor c/2 \rfloor, \qquad \|y + w\|_1 = c - \lfloor c/2 \rfloor = \lceil c/2 \rceil.$$

Since $a + b \ge 2$ and $a + b + c \le 2t$, both residual radii are nonnegative. They differ by at most one and satisfy $p' + q' = 2t - c \ge a + b$. The near-equal-radius identity (15) now makes the displayed local increment positive, including its $a - b = 2$ half-increment boundary. Proposition 4.2 expresses the full increment as a sum of nonnegative slice increments containing this positive term. □

Theorem 4.4 (Lee-lens majorization and active-shell strictness) *Fix $d \ge 2$ and $p, q \in \mathbb{Z}_{\ge 0}$. If $\lambda, \mu \in \mathbb{Z}^d_{\ge 0}$ have equal coordinate sum and $\lambda \succeq \mu$, then*

$$J^{(d)}_{p,q}(\lambda) \le J^{(d)}_{p,q}(\mu). \tag{19}$$

Thus every discrete cross-correlation of two Lee balls, including balls of different radii, is Schur-concave in the decreasing rearrangement of $(|u_1|, \ldots, |u_d|)$. Equivalently, the joint norm enumerators are ordered in bivariate lower-orthant order: every rectangle $\{\|x\|_1 \le p, \|x+u\|_1 \le q\}$ obeys the same majorization inequality. For equal radii $p = q = t$, if the common coordinate sum is at most $2t$, then the inequality is strict whenever λ and μ are not permutations of one another.

Proof The weak inequality is the Lee-ball specialization of Corollary 3.5. Equivalently, it follows by summing the exact residual kernel in Proposition 4.2 along a unit-transfer chain.

For equal radii in the active range, apply Lemma 4.3 to the two coordinates used by each nontrivial transfer. It supplies one strictly increasing slice, while Proposition 4.2 makes every other slice nondecreasing. Hence every nontrivial step, and therefore every strict majorization chain, is strict. □

4.2 Global stability and equality

Let $\mathcal{P}_d(s)$ be the decreasing nonnegative integer partitions of s into d parts. Join λ to μ when μ is obtained from λ by one unit balancing transfer followed by sorting. Give this directed edge the equal-radius weight

$$w_t(\lambda, \mu) = J^{(d)}_{t,t}(\mu) - J^{(d)}_{t,t}(\lambda).$$

The next definition fixes, once and for all, the shell terminology used for every body in this paper.

Definition 4.5 (Active shells) Let $d \ge 2$, let $A, B \subset \mathbb{Z}^d$ be finite signed-permutation-invariant sets whose fixed-shell unit-transfer arc weights are all nonnegative, and let $s \in \mathbb{Z}_{\ge 2}$. The shell s is *active* for (A, B) if the least unit-transfer arc weight

$$\min_{\lambda \to \mu \text{ in } \mathcal{P}_d(s)} \bigl(g_{A,B}(\mu) - g_{A,B}(\lambda)\bigr)$$

is strictly positive, that is, if every balancing transfer on the shell strictly increases the overlap. A shell on which this minimum vanishes is *inactive*.

Theorems 4.17, 5.1, 5.4, and 6.4 make this terminology exact for the solved families: the active shells are $2 \le s \le 2t$ for equal Lee balls and $2 \le s \le 2t+1$ for equal cubes, while for $C_3 + bB^3_\infty$ every shell $2 \le s \le 2b+3$ is active and every shell $s \ge 2b+4$ is inactive.

The next definition names the distinguished arc that attains the sharp constants of the Lee and cube families. Throughout the paper, the words *axial shift* and *axial orbit* are reserved for the concentrated shift se_1 and its signed-permutation orbit; the edge below is always referred to by its full name.

Definition 4.6 (Canonical axial-residual edge) For $d \ge 3$ and $s \ge 2$, the *canonical axial-residual edge* of $\mathcal{P}_d(s)$ is the unit-transfer arc

$$\mathrm{sort}(s-2, 2, 0^{d-2}) \longrightarrow \mathrm{sort}(s-2, 1, 1, 0^{d-3}),$$

in which the transfer acts on the pair $(2, 0)$ and the residual mass $s-2$ is supported on at most one coordinate. For $s = 2$ it is the arc $(2, 0, \ldots) \to (1, 1, 0, \ldots)$, and for $s = 3$ it is $(2, 1, 0, \ldots) \to (1, 1, 1, 0, \ldots)$.

For comparable partitions $\lambda \succeq \mu$, define

$$d_M(\lambda, \mu) = \frac{1}{2}\|\lambda - \mu\|_1. \tag{20}$$

The following self-contained lemma identifies d_M with the graph distance; it is a quantitative form of the classical fact that integer majorization is generated by unit transfers [44].

Lemma 4.7 (Exact transfer distance) *Let $\lambda \succeq \mu$ in $\mathcal{P}_d(s)$ with $\lambda \ne \mu$. Then the minimum number of unit balancing transfers, each followed by sorting, needed to pass from λ to μ equals $d_M(\lambda, \mu)$, and an optimal chain can be chosen so that every intermediate partition majorizes μ.*

Proof Lower bound. Let $\nu = \mathrm{sort}(\lambda - e_i + e_j)$ be one step. Sorting two vectors decreasingly does not increase their ℓ_1 distance, so $\|\nu - \lambda\|_1 \le \|(\lambda - e_i + e_j) - \lambda\|_1 = 2$, and the triangle inequality gives $\|\nu - \mu\|_1 \ge \|\lambda - \mu\|_1 - 2$. Hence each step decreases $d_M(\cdot, \mu)$ by at most one, and at least $d_M(\lambda, \mu)$ steps are needed.

Upper bound. Suppose $\lambda \ne \mu$. Let i be the first index with $\lambda_i \ne \mu_i$; since the prefix sums of λ dominate those of μ and agree before i, necessarily $\lambda_i > \mu_i$. Let $j > i$ be the first later index with $\lambda_j < \mu_j$; it exists because the total sums agree. Then $\lambda_i \ge \mu_i + 1 \ge \mu_j + 1 \ge \lambda_j + 2$, using that μ is decreasing, so the transfer $\nu = \mathrm{sort}(\lambda - e_i + e_j)$ is legal.

The step decreases the distance by exactly one. Indeed, coordinate i has $|\lambda_i - 1 - \mu_i| = \lambda_i - \mu_i - 1$ and coordinate j has $|\lambda_j + 1 - \mu_j| = \mu_j - \lambda_j - 1$, so $\|(\lambda - e_i + e_j) - \mu\|_1 = \|\lambda - \mu\|_1 - 2$; sorting does not increase the ℓ_1 distance to the sorted μ, and the lower-bound step above prevents a decrease by more than two.

The new partition still majorizes μ. For $h < i$ and $h \geq j$ the prefix sums of $\lambda - e_i + e_j$ are those of λ. For $i \leq h < j$ they decrease by one, but on this range the prefix excess of λ over μ is at least one: at $h = i$ it equals $\lambda_i - \mu_i \geq 1$, and for $i < h < j$ the choice of j gives $\lambda_k \geq \mu_k$ for $i < k \leq h$. Since sorting decreasingly only increases prefix sums, $\nu \succeq \mu$. Iterating $d_M(\lambda, \mu)$ times reaches ℓ_1 distance zero, that is, μ. □

Proposition 4.8 (Shell stability bookkeeping) *For $d \in \mathbb{Z}_{\geq 2}$, $t \in \mathbb{Z}_{\geq 1}$, and $s \in \{2, \ldots, 2t\}$, put*

$$\kappa_{d,t,s} = \min_{\lambda \to \mu \text{ in } \mathcal{P}_d(s)} w_t(\lambda, \mu). \tag{21}$$

Then $\kappa_{d,t,s}$ is a positive integer and, whenever $\lambda \succeq \mu$,

$$J^{(d)}_{t,t}(\mu) - J^{(d)}_{t,t}(\lambda) \geq \kappa_{d,t,s} d_M(\lambda, \mu) \geq d_M(\lambda, \mu). \tag{22}$$

Moreover,

$$\kappa_{d,t,s} = \min_{\substack{\lambda \succeq \mu,\ \lambda \neq \mu \\ \lambda, \mu \in \mathcal{P}_d(s)}} \frac{J^{(d)}_{t,t}(\mu) - J^{(d)}_{t,t}(\lambda)}{d_M(\lambda, \mu)}. \tag{23}$$

Hence $\kappa_{d,t,s}$ is the largest constant for which (22) holds simultaneously for every comparable pair on the shell.

Proof The graph is finite. The strict clause of Theorem 4.4 makes every edge weight positive in the active range $s \leq 2t$, hence $\kappa_{d,t,s} \geq 1$. By Lemma 4.7, a shortest transfer path from λ to μ has $d_M(\lambda, \mu)$ edges. Telescoping the lens counts along it and bounding every edge by (21) proves (22). Conversely, an edge attaining the minimum has $d_M = 1$, so no larger constant can hold for all comparable pairs. This proves (23). □

Remark (Scope of Proposition 4.8) Given positivity of the edge weights (Theorem 4.4) and the identification of graph distance with d_M (Lemma 4.7), both (22) and (23) are formal. What has to be done is to evaluate κ and its extremizers, in Theorems 4.17, 4.25, 4.26, and 6.4.

Corollary 4.9 (Complete chain equality certificate) *Let $p, q \in \mathbb{Z}_{\geq 0}$ and let $\lambda = \lambda^{(0)} \to \cdots \to \lambda^{(m)} = \mu$ be any balancing chain. Then*

$$J^{(d)}_{p,q}(\lambda) = J^{(d)}_{p,q}(\mu)$$

if and only if every edge of the chain satisfies the residual parity condition of Proposition 4.2. Equivalently, all positive residual histogram entries are supported on the two-channel zero set (14). In the equal-radius active range this occurs only for $m = 0$; hence a comparable pair $\lambda \succeq \mu$ has equal lens values if and only if $\lambda = \mu$.

Proof The endpoint difference is the sum of the nonnegative edge increments. It vanishes exactly when each edge increment vanishes, and Proposition 4.2 gives an iff criterion for each one. The last assertion follows from positivity in Proposition 4.8. □

Remark 4.10 (Order-theoretic scope of strictness) The equality statements above are relative to the majorization order: strict Schur-concavity controls comparable pairs and does not force the lens to separate incomparable orbits. Already on the outermost active shell of Fig. 8, namely $\mathcal{P}_3(6)$ with $t = 3$, the compositions $\lambda = (4,1,1)$ and $\mu = (3,3,0)$ are incomparable ($4 > 3$ but $4 + 1 < 3 + 3$) and are not signed permutations of one another, yet

$$J^{(3)}_{3,3}(4,1,1) = 4 = J^{(3)}_{3,3}(3,3,0) :$$

the four lattice points in the first lens are $(-3,0,0)$, $(-2,-1,0)$, $(-2,0,-1)$, $(-1,-1,-1)$, and in the second $(-3,0,0)$, $(-2,-1,0)$, $(-1,-2,0)$, $(0,-3,0)$. Coincidences of this kind between incomparable compositions are compatible with every theorem in this paper.

4.3 The middle-section evaluation

The graph definition (21) is now evaluated by a discrete middle-section principle. Extend the Lee counts by the uniform conventions

$$L_j(r) = 0 \quad (r < 0), \qquad L_0(r) = 1 \quad (r \in \mathbb{Z}_{\geq 0}), \qquad \binom{K+e}{e} = 0 \quad (K < 0),$$

which cover every boundary value below. The principle is assembled from four short lemmas: a coordinatewise representation of the bisector layers, a signed Erdős–Ko–Rado incidence bound, an incidence-to-count conversion, and an axial equality case.

Remark (Equality and strictness) The proof of Theorem 4.17 passes through exactly two inequalities, recorded here once. Fix an edge of $\mathcal{P}_d(s)$ with residual shift $w \in \mathbb{Z}^m$, $m = d-2$, $c = \|w\|_1$, and $H = 2t - s + 1$; let A_k be the number of central residual points of excess k and $\eta \in \{1/2, 1\}$ the kernel factor of that proof.

Link 1: noncentral slices are dropped. Retaining in the residual-convolution sum of Proposition 4.2 only the slices with $|r - r'| \leq 1$ gives $w_t(\lambda,\mu) \geq \sum_{k\geq 0} \eta A_k (H-2k)_+$, termwise since the discarded weights are nonnegative, with equality exactly when every noncentral slice weight vanishes. For the canonical axial-residual edge the kernel table $\Phi_{p,q}(2,2)$ in the proof is zero whenever $|p-q| \geq 2$.

Link 2: central counts are replaced by the universal model. Lemmas 4.11–4.14 give the cumulative bound $\sum_{k\leq K} \eta A_k \geq L_{m-1}(K)$ for $c > 0$, and for $c = 0$ every residual point is central, with cumulative count $L_m(K) \geq L_{m-1}(K)$. Against the nonincreasing weights $(H-2k)_+$, Lemma 4.16 turns this into $\sum_k \eta A_k (H-2k)_+ \geq \sum_k S_{m-1}(k)(H-2k)_+$, whose closed evaluation (29) no longer depends on the edge. By Abel summation, equality holds as soon as the cumulative bound is an equality for every K with $(H-2K)_+ > (H-2K-2)_+$, which Lemma 4.15 verifies for a concentrated residual ce_1 with $c \geq 2$.

The minimum therefore requires equality in both links, and the canonical axial-residual edge meets both. All edges meeting both are determined in Theorem 4.25.

Lemma 4.11 (Middle-layer representation) *Let $m \geq 1$ and $w \in \mathbb{Z}^m_{\geq 0}$ with $c = \|w\|_1 > 0$; let h be the number of positive coordinates of w and $z = m - h$. For $y \in \mathbb{Z}^m$ put*

$$r(y) = \|y\|_1, \qquad r'(y) = \|y + w\|_1, \qquad k(y) = \frac{r(y) + r'(y) - c}{2},$$

and, for $1 \le i \le h$,

$$q_i = \operatorname{clamp}(-y_i, 0, w_i), \qquad \ell_i = \operatorname{dist}(-y_i, [0, w_i]),$$

where $\operatorname{clamp}(x, 0, w_i) = \min\{w_i, \max\{0, x\}\}$*. Then, coordinatewise,*

$$|y_i| - |y_i + w_i| = 2q_i - w_i, \qquad |y_i| + |y_i + w_i| = w_i + 2\ell_i. \tag{24}$$

Consequently $r(y) = r'(y)$ *if and only if* $\sum_i q_i = c/2$ *(c even), and* $|r(y) - r'(y)| = 1$ *if and only if* $\sum_i q_i \in \{(c-1)/2, (c+1)/2\}$ *(c odd); in these cases*

$$k(y) = \sum_{i=1}^{h} \ell_i + \sum_{i>h} |y_i| \in \mathbb{Z}_{\ge 0}.$$

Moreover, fix a point q *of the integer box* $\prod_{i\le h}[0, w_i]$ *and let* $e(q)$ *be the number of its coordinates lying at an endpoint of its interval. The points* y *realizing* q *are parametrized by one independent outward tail* $\ell_i \in \mathbb{Z}_{\ge 0}$ *for each endpoint coordinate together with the values* $v \in \mathbb{Z}^z$ *of the zero coordinates, and for each fixed* v *the number of tails of total length at most* K *is* $\binom{K+e(q)}{e(q)}$.

Proof If $-y_i \in [0, w_i]$, then $|y_i| = q_i$, $|y_i + w_i| = w_i - q_i$, and $\ell_i = 0$. If $-y_i > w_i$, then $q_i = w_i$, $\ell_i = -y_i - w_i$, $|y_i| = w_i + \ell_i$, and $|y_i + w_i| = \ell_i$. If $-y_i < 0$, then $q_i = 0$, $\ell_i = y_i$, $|y_i| = \ell_i$, and $|y_i + w_i| = w_i + \ell_i$. This proves (24). Summing the first identity over $i \le h$ and noting that a zero coordinate contributes $|y_i| - |y_i + w_i| = 0$ shows that $r(y) - r'(y) = 2\sum_i q_i - c$, which gives both layer characterizations; summing the second identity gives the excess formula, since a zero coordinate contributes $2|y_i|$. A coordinate with $0 < q_i < w_i$ forces $\ell_i = 0$ and $y_i = -q_i$, while a coordinate with $q_i \in \{0, w_i\}$ admits exactly one outward direction with a free tail length $\ell_i \in \mathbb{Z}_{\ge 0}$. The tails therefore form a free tuple in $\mathbb{Z}_{\ge 0}^{e(q)}$, and the nonnegative integer solutions of $\ell_1 + \cdots + \ell_{e(q)} \le K$ number $\binom{K+e(q)}{e(q)}$. □

Lemma 4.12 (Face–layer incidence bound) *Let* $w_1, \dots, w_h > 0$ *be integers with sum* c*. Encode each codimension-*j *face of the box* $\prod_i [0, w_i]$ *by a signed* j*-set* (P, N)*: the coordinates in* P *are fixed at their upper endpoints* w_i *and those in* N *at zero, with* $P \cap N = \varnothing$ *and* $|P| + |N| = j$*. For* $0 \le j \le h$ *there are* $2^j\binom{h}{j}$ *such faces. If* $c = 2a$ *is even, at least*

$$2^j \binom{h-1}{j}$$

of them meet the layer $\Lambda_a = \{q : \sum_i q_i = a\}$*; see Figure 7. If* $c = 2a + 1$ *is odd, the number of incidences between codimension-*j *faces and the two layers* Λ_a *and* Λ_{a+1} *is at least*

$$2^{j+1} \binom{h-1}{j}.$$

Proof Write $w(P) = \sum_{i\in P} w_i$ and $w(N) = \sum_{i\in N} w_i$. On the face (P, N) the coordinate sum ranges over the integer interval $[w(P),\, c - w(N)]$, and the free coordinates change it in unit steps, so the face contains a point of the integer layer Λ_a if and only if $w(P) \le a \le c - w(N)$.

Let $c = 2a$. The face misses Λ_a exactly when $w(P) > a$ or $w(N) > a$; these events are mutually exclusive because $w(P) + w(N) \le c = 2a$. Consider the family of misses with $w(P) > a$. If two of its members had disjoint plus supports, $P_1 \cap P_2 = \varnothing$ would give $w(P_1) + w(P_2) \le c = 2a$, contradicting $w(P_1), w(P_2) > a$. Hence any two members share a coordinate signed plus in both, so the family is intersecting as a family of signed j-sets. The signed Erdős–Ko–Rado theorem of Bollobás and Leader [65] bounds every intersecting family of signed j-sets on h coordinates by $2^{j-1}\binom{h-1}{j-1}$; the same argument applies to the misses with $w(N) > a$ through their minus supports. Therefore at least

$$2^j\binom{h}{j} - 2\cdot 2^{j-1}\binom{h-1}{j-1} = 2^j\left[\binom{h}{j} - \binom{h-1}{j-1}\right] = 2^j\binom{h-1}{j}$$

faces meet the middle layer. For $j = 0$ there is nothing to subtract: the single face is the whole box and meets the layer.

Let $c = 2a+1$. For Λ_a the miss conditions are $w(P) \ge a+1$ or $w(N) \ge a+2$; for Λ_{a+1} they are $w(P) \ge a+2$ or $w(N) \ge a+1$. Every listed support has weight exceeding $c/2$, so two disjoint supports would exceed c in total, and each of the four exceptional families is signed-intersecting exactly as before, hence of size at most $2^{j-1}\binom{h-1}{j-1}$. Among the $2 \cdot 2^j\binom{h}{j}$ face–layer incidences at least

$$2^{j+1}\binom{h}{j} - 4\cdot 2^{j-1}\binom{h-1}{j-1} = 2^{j+1}\binom{h-1}{j}$$

remain. ☐

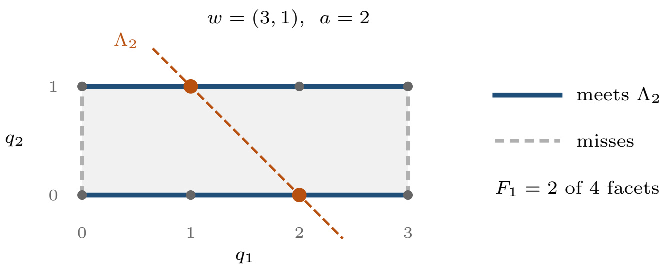

Fig. 7 The incidence bound of Lemma 4.12 in its smallest nontrivial instance, $h = 2$ and $w = (3, 1)$, so $c = 4$ and $a = 2$. A facet is missed exactly when the coordinate it freezes is too heavy, here the two facets $q_1 = 0$ and $q_1 = 3$; the remaining two meet Λ_2, which matches the bound $2^1\binom{h-1}{1} = 2$. Neither vertex meets the layer, again matching $2^2\binom{h-1}{2} = 0$.

Lemma 4.13 (Incidence-to-count conversion) *In the setting of Lemma 4.11, fix $K \in \mathbb{Z}_{\ge 0}$ and an integer layer Λ_a with $0 \le a \le c$, and let $F_j(a)$ be the number of codimension-j faces of $\prod_{i \le h}[0, w_i]$ meeting Λ_a. Then*

$$\sum_{q \in \Lambda_a}\binom{K + e(q)}{e(q)} \ge \sum_{j \ge 0} F_j(a)\binom{K}{j}.$$

Moreover, for every $j \ge 0$,

$$\sum_{v \in \mathbb{Z}^z} L_j\big(K - \|v\|_1\big) = L_{j+z}(K). \tag{25}$$

Proof Vandermonde's identity gives

$$\binom{K+e(q)}{e(q)} = \sum_{j=0}^{e(q)} \binom{e(q)}{j}\binom{K}{j}. \tag{26}$$

The binomial $\binom{e(q)}{j}$ is exactly the number of codimension-j faces containing q: such a face must fix j of the $e(q)$ endpoint coordinates of q, and the sign of each fixed coordinate is determined by which endpoint that coordinate occupies. Summing over $q \in \Lambda_a$ therefore counts the face–point incidences of the layer. Every face meeting the layer contains at least one of its lattice points, so $\sum_{q\in\Lambda_a} \binom{e(q)}{j} \geq F_j(a)$, and substituting into (26) proves the first claim. For (25), both sides count the set $\{(x,v) \in \mathbb{Z}^j \times \mathbb{Z}^z : \|x\|_1 + \|v\|_1 \leq K\}$: the left-hand side groups its points by v, and the right-hand side is the Lee ball of radius K in $j+z$ coordinates. □

Lemma 4.14 (Central Lee-bisector sections of integer boxes) *Let $m \geq 1$, $w \in \mathbb{Z}^m_{\geq 0}$, and $c = \|w\|_1 > 0$. For $K \in \mathbb{Z}_{\geq 0}$, let $N_w(K)$ count the points $y \in \mathbb{Z}^m$ with $k(y) \leq K$ and*

$$r(y) = r'(y) \quad (c \text{ even}), \qquad |r(y) - r'(y)| = 1 \quad (c \text{ odd}).$$

Then

$$N_w(K) \geq \begin{cases} L_{m-1}(K), & c \text{ even}, \\ 2L_{m-1}(K), & c \text{ odd}. \end{cases} \tag{27}$$

Proof By Lemma 4.11, the counted points are parametrized by a layer point q (of $\Lambda_{c/2}$ when c is even, of $\Lambda_{(c\pm1)/2}$ when c is odd), a value $v \in \mathbb{Z}^z$ of the zero coordinates, and outward tails of total length at most $K - \|v\|_1$. Hence

$$N_w(K) = \sum_{\text{layers}} \sum_{v\in\mathbb{Z}^z} \sum_{q} \binom{(K-\|v\|_1)+e(q)}{e(q)}.$$

For each fixed v with $\|v\|_1 \leq K$, Lemma 4.13 followed by Lemma 4.12 and the count (9) bounds the inner sum below by $L_{h-1}(K - \|v\|_1)$ in the even case and, after adding the two layers, by $2L_{h-1}(K - \|v\|_1)$ in the odd case. Summing over v with (25) gives $L_{h-1+z}(K) = L_{m-1}(K)$ and twice that value, respectively. □

Lemma 4.15 (Axial equality) *If $c \geq 2$ and $w = (c, 0, \ldots, 0)$, then equality holds in (27) for every $K \in \mathbb{Z}_{\geq 0}$.*

Proof Here $h = 1$ and $z = m - 1$. If $c = 2a$ is even, the layer $q_1 = a$ consists of the single point a, which is interior to $[0, c]$ because $c \geq 2$; it has $e = 0$, so no outward tail is allowed and $y_1 = -a$ is forced. If $c = 2a + 1$ is odd, then $c \geq 3$ and the two layers $q_1 \in \{a, a+1\}$ are again interior points with $e = 0$. In either case the excess $k(y)$ reduces to the Lee norm of the $m-1$ zero coordinates, so $N_w(K)$ equals exactly $L_{m-1}(K)$, respectively $2L_{m-1}(K)$. □

The passage from cumulative section counts to weighted edge bounds uses one further elementary lemma, stated separately because it carries the whole majorization-of-sequences step.

Lemma 4.16 (Cumulative dominance) *Let $(a_k)_{k\ge0}$ and $(b_k)_{k\ge0}$ be nonnegative sequences with*

$$\sum_{k=0}^{K} a_k \ge \sum_{k=0}^{K} b_k \qquad \text{for every } K \in \mathbb{Z}_{\ge0},$$

and let $(w_k)_{k\ge0}$ be nonnegative, nonincreasing, and eventually zero. Then

$$\sum_{k\ge0} a_k w_k \ge \sum_{k\ge0} b_k w_k.$$

Proof Choose M with $w_k = 0$ for all $k > M$, and put $A_K = \sum_{k\le K} a_k$ and $B_K = \sum_{k\le K} b_k$. Abel summation gives

$$\sum_{k=0}^{M} a_k w_k = \sum_{K=0}^{M} A_K (w_K - w_{K+1}), \qquad \text{with } w_{M+1} = 0,$$

and likewise for (b_k). Every factor $w_K - w_{K+1}$ is nonnegative, so $A_K \ge B_K$ termwise gives the claim. □

Theorem 4.17 (Sharp active-shell constant) *For $d \in \mathbb{Z}_{\ge2}$, $t \in \mathbb{Z}_{\ge1}$, and $s \in \{2,\dots,2t\}$, the following formulas hold. If $d = 2$, then*

$$\kappa_{2,t,s} = 2t - s + 1. \tag{28}$$

If $d \ge 3$, put $T_{t,s} = t - \lceil s/2 \rceil$. Then

$$\kappa_{d,t,s} = \begin{cases} L_{d-1}(t-1), & s = 2,\\ \big(L_{d-1}(t-1) - L_{d-3}(t-1)\big)/2, & s = 3,\\ L_{d-2}(T_{t,s}), & s \ge 4 \text{ even},\\ L_{d-2}(T_{t,s}) + L_{d-3}(T_{t,s}), & s \ge 5 \text{ odd}. \end{cases} \tag{29}$$

For $d \ge 3$, the minimum on every active shell is attained by the canonical axial-residual edge of Definition 4.6, where 0^j denotes j trailing zeros:

$$\operatorname{sort}(s-2, 2, 0^{d-2}) \longrightarrow \operatorname{sort}(s-2, 1, 1, 0^{d-3}). \tag{30}$$

The cases $s = 2$ and $s = 3$ are evaluated separately in the proof ($s = 2$ has a single arc, and for $s = 3$ the two arc weights are compared exactly), while $s \ge 4$ follows from the axial-section argument. The complete list of minimizing arcs, on every active shell, is Theorem 4.25. Finally, extend the definition (21) to all $s \in \mathbb{Z}_{\ge2}$. Then

$$\kappa_{d,t,s} = 0 \qquad (s > 2t), \tag{31}$$

so the active shells of two equal Lee balls are exactly $2 \le s \le 2t$.

Proof The two-dimensional formula follows immediately from the equal-radius clause of Theorem 4.1, because every edge on $\mathcal{P}_2(s)$ has $R = s$.

Let $d \ge 3$ and consider an arbitrary edge represented before sorting by

$$(a, b, w) \longrightarrow (a-1, b+1, w), \qquad R = a + b, \qquad D = a - b \ge 2.$$

Put $m = d - 2$, $c = \|w\|_1 = s - R$, and $H = 2t - s + 1$. On a central residual point of excess k, the remaining radii $p = t - r(y)$ and $q = t - r'(y)$ differ by at most one and satisfy

$$p + q - R + 1 = H - 2k. \tag{32}$$

Whenever $H - 2k > 0$, both radii are nonnegative because $R \geq 2$; otherwise the claimed lower-bound summand is zero. Thus (15) gives the local weight $(H - 2k)_+$, except when c is odd and $D = 2$, when it gives half that value. Let A_k be the number of central residual points of excess k, and put $\eta = 1/2$ in this exceptional case and $\eta = 1$ otherwise. The two lines of (27) imply in every case with $c > 0$ that

$$\sum_{k=0}^{K} \eta A_k \geq L_{m-1}(K). \tag{33}$$

Indeed, the factor two in the odd bound cancels the sole half-increment; when $D > 2$ the odd bound is stronger than needed. If $c = 0$, every residual point is central and the cumulative count is $L_m(K) \geq L_{m-1}(K)$. All noncentral summands in Proposition 4.2 are nonnegative.

Let $S_j(k) = L_j(k) - L_j(k-1)$. Apply Lemma 4.16 with $a_k = \eta A_k$, $b_k = S_{m-1}(k)$, and the nonincreasing, eventually zero weight sequence $w_k = (H - 2k)_+$; the cumulative hypothesis is (33) for $c > 0$ and the bound $\sum_{k \leq K} A_k = L_m(K) \geq L_{m-1}(K)$ for $c = 0$. This gives, for every edge,

$$w_t(\lambda, \mu) \geq \sum_{k \geq 0} S_{m-1}(k)(H - 2k)_+. \tag{34}$$

If s is even, write $H = 2T + 1$ with $T = t - s/2$. For each $z \in \mathbb{Z}^{m-1}$ of norm $k \leq T$, the factor $2(T - k) + 1$ counts the choices of one further integer coordinate. Hence the right-hand side of (34) is $L_m(T)$. If s is odd, write $H = 2(T+1)$ with $T = t - (s+1)/2$. Now the factor $2(T-k)+2$ is the sum of the same one-coordinate count and one extra choice, giving $L_m(T) + L_{m-1}(T)$.

For $s \geq 4$, the edge (30) has transfer pair $(2, 0)$ and residual shift $(s - 2, 0, \ldots, 0)$. Lemma 4.15 then turns the central-section bound (27) into an equality. Equivalently, the one-coordinate series in Theorem B.1 (Appendix B) and lower-orthant summation give the atomic identity

$$\sum_{p,q \geq 0} \Phi_{p,q}(2,2) X^p Y^q = \frac{XY(1+X)(1+Y)}{(1 - XY)^2}.$$

Coefficient extraction, or direct specialization of (13), gives

$$\Phi_{p,q}(2,2) = \begin{cases} (2p-1)_+, & p = q, \\ \min\{p,q\}, & |p - q| = 1, \\ 0, & |p - q| \geq 2, \end{cases}$$

so no noncentral residual slice adds weight. The lower bounds above are therefore attained, proving the last two lines of (29).

For $s = 2$, the shell graph has the single edge $(2, 0^{d-1}) \to (1, 1, 0^{d-2})$. Its weight is

$$\sum_{y \in \mathbb{Z}^{d-2}} \bigl(2(t - \|y\|_1) - 1\bigr)_+ = L_{d-1}(t-1),$$

because the linear factor counts one additional integer coordinate. For $s = 3$, put $m = d-2$ and $T = t - 2$; the shell graph has exactly two edges. Consider first $(2, 1, 0^{d-2}) \to (1, 1, 1, 0^{d-3})$. Its transfer moves one unit from the coordinate at two into a zero coordinate, so the transfer pair is $(2, 0)$ with $R = D = 2$, and the residual shift is $w = e_1 \in \mathbb{Z}^m$. Write a residual point as $y = (n, z)$ with $n \in \mathbb{Z}$ and $z \in \mathbb{Z}^{m-1}$; then

$$r = \|y\|_1 = |n| + \|z\|_1, \qquad r' = \|y + e_1\|_1 = |n+1| + \|z\|_1,$$

so every residual slice has $|r-r'|=1$, and its remaining radii $p=t-r$ and $q=t-r'$ differ by one. By the displayed table, the slice weight is $\min\{p,q\}$, which together with the zero extension for negative radii equals $\bigl(t-\|z\|_1-\max\{|n|,|n+1|\}\bigr)_+$. As n runs over $\mathbb{Z}$, the quantity $\max\{|n|,|n+1|\}$ takes every positive integer value exactly twice, so Proposition 4.2 evaluates the edge weight as

$$B_T = 2\sum_{z\in\mathbb{Z}^{m-1}}\sum_{n\geq 0}(T+1-n-\|z\|_1)_+. \tag{35}$$

Multiplying the three series $\sum_z X^{\|z\|_1} = \bigl((1+X)/(1-X)\bigr)^{m-1}$, $\sum_{n\geq 0} X^n = 1/(1-X)$, and $\sum_{T\geq 0}(T+1)X^T = 1/(1-X)^2$ gives

$$\sum_{T\geq 0} B_T X^T = \frac{2(1+X)^{m-1}}{(1-X)^{m+2}} = \frac{1}{2X}\left[\frac{(1+X)^{m+1}}{(1-X)^{m+2}} - \frac{(1+X)^{m-1}}{(1-X)^{m}}\right], \tag{36}$$

where the bracket identity follows from $(1+X)^2-(1-X)^2=4X$. Since $\sum_{r\geq 0} L_j(r)X^r = (1+X)^j/(1-X)^{j+1}$ and the bracket has zero constant term, comparing coefficients in (36) gives

$$B_T = \tfrac{1}{2}\bigl(L_{m+1}(T+1)-L_{m-1}(T+1)\bigr) = \tfrac{1}{2}\bigl(L_{d-1}(t-1)-L_{d-3}(t-1)\bigr).$$

The only other edge on the shell is $(3,0^{d-1})\to(2,1,0^{d-2})$, with transfer pair $(3,0)$, hence $R=D=3$, and zero residual shift. Every residual slice has $r=r'=\|y\|_1$ and equal remaining radii $p=q=t-\|y\|_1$, so the equal-radius clause of Theorem 4.1 gives the slice weight $\Phi_{p,p}(3,3)=(2p-2)_+$, and the edge weight is

$$A_T = \sum_{y\in\mathbb{Z}^m}\bigl(2(t-\|y\|_1)-2\bigr)_+ = 2\sum_{y\in\mathbb{Z}^m}(T+1-\|y\|_1)_+.$$

The same series give

$$\sum_{T\geq 0} A_T X^T = \frac{2(1+X)^m}{(1-X)^{m+2}} = (1+X)\sum_{T\geq 0} B_T X^T.$$

Consequently $A_T = B_T + B_{T-1} \geq B_T$, with $B_{-1}=0$, proving the $s=3$ line; the minimum is attained by the first edge, which is the canonical axial-residual edge of $\mathcal{P}_d(3)$. For $s=2$ the single arc computed above is the canonical axial-residual edge of $\mathcal{P}_d(2)$, so the attainment claim holds on every active shell.

For $s>2t$ and $\|u\|_1=s$, any x counted by $J^{(d)}_{t,t}(u)$ would give

$$s=\|u\|_1 \leq \|x\|_1 + \|x+u\|_1 \leq 2t,$$

a contradiction. Hence every lens value on the shell vanishes, every arc weight is zero, and (31) follows, completing the proof. □

Corollary 4.18 (Explicit sharp deficit) *For every comparable $\lambda\succeq\mu$ in $\mathcal{P}_d(s)$, the right-hand side of (22) remains valid with $\kappa_{d,t,s}$ replaced by the corresponding closed expression in (28) or (29), and no larger uniform coefficient is possible. In particular,*

$$\kappa_{d,t,2t}=1, \qquad \kappa_{d,t,2t-1}=2 \quad (t\geq 2).$$

Figure 8 shows one complete shell graph with the exact arc weights that Proposition 4.8 sums and Theorem 4.17 minimizes.

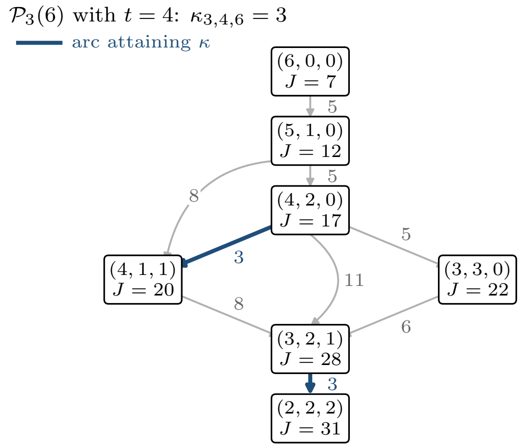

Fig. 8 The unit-transfer graph $\mathcal{P}_3(6)$ for $t = 4$. Vertices are the partitions of six into at most three parts, labelled by their lens values $J = J^{(3)}_{4,4}$; each arrow is one legal balancing transfer, labelled by its exact increment. All nine weights are positive, which gives stability along any path, and the smallest of them is $\kappa_{3,4,6} = 3$, attained exactly on the two highlighted arcs.

4.4 Classification of minimizing arcs

Theorem 4.17 evaluates the sharp constant and exhibits one attaining arc. This subsection determines *every* arc attaining $\kappa_{d,t,s}$, on every active shell, and then upgrades the arc classification to a characterization of all *pairs* attaining equality in the stability inequality (22). The classification carries a rigidity that the constant alone does not show: on all interior shells the minimizers form one explicit family whose size depends only on s—neither on the dimension nor on the radius—and no two of its members concatenate into a distance-two extremal configuration, so equality in (22) is confined to single arcs. The two outermost shells acquire an additional, also completely described, boundary family, along which arbitrarily long extremal geodesics do exist. Throughout, an arc of $\mathcal{P}_d(s)$ is represented modulo signed permutations by its transfer pair (a, b) with $a \geq b + 2$ and its residual $w \in \mathbb{Z}^{d-2}_{\geq 0}$, and we keep the proof notation $m = d - 2$, $R = a + b$, $D = a - b$, $c = \|w\|_1 = s - R$, and $H = 2t - s + 1$. An active shell of weight s is called *interior* when $d \geq 3$ and $4 \leq s \leq 2t - 2$, equivalently $H \geq 3$; the two *outermost* shells are $s = 2t - 1$ ($H = 2$) and $s = 2t$ ($H = 1$). This is the sense in which "interior shell" is used throughout the paper, including the abstract and introduction.

Definition 4.19 (Balanced-gap axial family) For $d \geq 3$ and $s \geq 4$, the *balanced-gap axial family* of $\mathcal{P}_d(s)$ consists of the arcs

$$\mathrm{sort}(c, b+2, b, 0^{d-3}) \longrightarrow \mathrm{sort}(c, b+1, b+1, 0^{d-3}), \qquad 0 \leq b \leq \tfrac{s-4}{2}, \quad c = s - 2b - 2, \tag{37}$$

that is, all unit transfers whose donor–recipient gap is exactly two and whose residual mass $c \geq 2$ is concentrated in a single coordinate. The member $b = 0$ is the canonical axial-residual edge of Definition 4.6, and the family contains exactly $\lfloor s/2 \rfloor - 1$ arcs.

The classification proceeds by a partition of all arcs into four shapes, displayed in Figure 9. We record the partition as a lemma so that exhaustiveness is checkable.

Lemma 4.20 (Case partition) *Every arc of $\mathcal{P}_d(s)$, represented modulo signed permutations by its transfer pair (a, b) with $a \geq b + 2$ and its residual $w \in \mathbb{Z}_{\geq 0}^{d-2}$, satisfies exactly one of the following, where $D = a - b$ and $c = \|w\|_1$:*

(I) *$D = 2$ and $w = ce_1$ with $c \geq 2$* (family shape)*;*

(II) *$c = 0$, or $D = 2$ and $w = e_1$* (outer arcs)*;*

(III) *w has at least two nonzero coordinates* (spread residuals)*;*

(IV) *$D \geq 3$ and $w = ce_1$ with $c \geq 1$* (wide-gap concentrated residuals).

Proof After permuting residual coordinates, either w has two or more nonzero coordinates, which is Case III and excludes the concentrated shapes of Cases I, II, and IV, or $w = ce_1$ for some $c \geq 0$. In the concentrated situation the pair (D, c) with $D \geq 2$ decides membership: $c = 0$ lies in Case II only; $c = 1$ lies in Case II if $D = 2$ and in Case IV if $D \geq 3$; and $c \geq 2$ lies in Case I if $D = 2$ and in Case IV if $D \geq 3$. These subranges of (D, c) are pairwise disjoint and cover all concentrated arcs. □

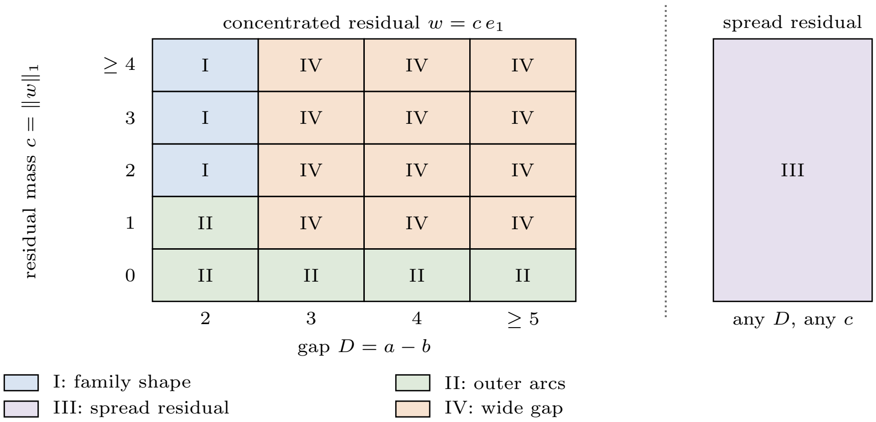

Fig. 9 The partition of Lemma 4.20 in the invariants $D = a - b$ and $c = \|w\|_1$. Concentrated residuals $w = ce_1$ occupy the left panel, where the pair (D, c) alone decides the case; residuals with two or more nonzero coordinates form Case III regardless of (D, c), shown on the right. Only Case I supports the sharp family.

Four further lemmas isolate the mechanisms. The first shows that gap-two transfers never collect noncentral weight, whatever the residual does.

Lemma 4.21 (Noncentral vanishing at gap two) *If $|p - q| \geq 2$, then $\Phi_{p,q}(R, 2) = 0$ for every $R \geq 2$.*

Proof Recall the clipped intervals $I(c') = [-p, p] \cap [-c' - q, -c' + q]$ from (16). If $q \geq p + 2$, then $I(0)$ and $I(2)$ both equal $[-p, p]$, so each parity bracket in (13) vanishes. If $p \geq q + 2$, then $I(0) = [-q, q]$ and $I(2) = [-q - 2, q - 2]$ both lie inside $[-p, p]$, and $I(2) = I(0) - 2$ is a parity-preserving translation, so again every parity count is unchanged. $\square$

The second lemma evaluates, exactly, the two arc shapes that will turn out to be minimizing only near the support boundary. Both weights are independent of data one might expect them to depend on.

Lemma 4.22 (Exact outer arc weights) *Let $d \geq 3$, $t \geq 1$, $4 \leq s \leq 2t$, $m = d - 2$, and $H = 2t - s + 1$.*

(i) *Every arc with vanishing residual, that is, every arc whose source and target are supported on the two transfer coordinates, has weight*

$$W_0 = \sum_{k \geq 0} S_m(k)(H - 2k)_+,$$

independently of its gap D.

(ii) *If $s \geq 5$ is odd, the gap-two arc with residual $w = e_1$, namely (37) with $b = (s-3)/2$ and $c = 1$, has weight*

$$W_1 = \sum_{k \geq 0} L_{m-1}(k)(H - 2k)_+.$$

(iii) *Both weights satisfy $W_0, W_1 \geq \kappa_{d,t,s}$, with equality if and only if $H \leq 2$.*

Proof (i) Here $w = 0$, so every residual point y has $r(y) = r'(y) = k$ and equal remaining radii $p = q = t - k$. For any $D \geq 2$ the full branch of (15) gives the slice weight $(2p - R + 1)_+ = (H - 2k)_+$, and the number of slices at norm k is $S_m(k)$.

(ii) As in the $s = 3$ computation inside the proof of Theorem 4.17, write a residual point as $y = (n, z)$ with $n \in \mathbb{Z}$, $z \in \mathbb{Z}^{m-1}$; then $|r - r'| = 1$ on every slice, the half branch of (15) applies, and the slice weight is $\frac{1}{2}(H - 2j)_+$ with $j = \min\{|n|, |n+1|\} + \|z\|_1$. Since $\min\{|n|, |n+1|\}$ takes each value $i \geq 0$ exactly twice, the number of slices with j fixed is $2\sum_{i \leq j} S_{m-1}(j - i) = 2L_{m-1}(j)$, and the halves cancel the factor two.

(iii) By the evaluation of (34), $\kappa_{d,t,s} = \sum_k S_{m-1}(k)(H - 2k)_+$ for $s \geq 4$. Hence

$$W_0 - \kappa_{d,t,s} = \sum_{k \geq 1} \big(S_m(k) - S_{m-1}(k)\big)(H - 2k)_+, \qquad W_1 - \kappa_{d,t,s} = \sum_{k \geq 1} L_{m-1}(k-1)(H - 2k)_+,$$

using $L_{m-1}(k) - S_{m-1}(k) = L_{m-1}(k-1)$. Every summand is nonnegative; for $k \geq 1$ the factors $S_m(k) - S_{m-1}(k)$ (points with nonzero last coordinate) and $L_{m-1}(k-1)$ are strictly positive, so each difference vanishes if and only if $(H - 2k)_+ = 0$ for all $k \geq 1$, that is, $H \leq 2$. $\square$

Lemma 4.23 (Spread residuals are strict) *Let $d \geq 4$, $4 \leq s \leq 2t$, and consider an arc whose residual w has at least two nonzero coordinates. Then its weight strictly exceeds $\kappa_{d,t,s}$.*

Proof Normalize signs so that $w \in \mathbb{Z}^m_{\geq 0}$, and let $N(j)$ count the points of the box $\prod_i [0, w_i]$ with coordinate sum j; these are exactly the residual points of excess zero at the corresponding split of c. We first claim $N(j) \geq 2$ for $1 \leq j \leq c-1$. Take any box point r with coordinate sum j. If there are indices $i \neq k$ with $r_i < w_i$ and $r_k > 0$, moving one unit from k to i gives a second point. Otherwise pick i_1 with $r_{i_1} > 0$; then $r_i = w_i$ for all $i \neq i_1$, and picking $i_2 \neq i_1$ with $w_{i_2} > 0$ forces, by the same argument with the roles reversed, $r_i = w_i$ for all $i \neq i_2$ as well, whence $r = w$ and $j = c$, a contradiction.

Now count central residual points of excess zero. If c is even, they are the box points at coordinate sum $c/2$, so $A_0 = N(c/2) \geq 2$, and $\eta = 1$ because the half branch requires odd c. If c is odd, then $c \geq 3$ because two coordinates are nonzero, and $A_0 = N\big((c-1)/2\big) + N\big((c+1)/2\big) \geq 4$, while $\eta \geq 1/2$. In all cases $\eta A_0 \geq 2 > 1 = L_{m-1}(0)$. Abel summation as in Lemma 4.16, with the cumulative bounds (33) at every K and the strict gain $(\eta A_0 - 1)\big(w_0 - w_1\big) \geq \min\{H, 2\} > 0$ at $K = 0$, shows that the central contribution alone exceeds $\sum_k S_{m-1}(k)(H-2k)_+ = \kappa_{d,t,s}$. □

Lemma 4.24 (Wide-gap witness) *Let $4 \leq s \leq 2t$, let the residual be concentrated, $w = ce_1$ with $c \geq 2$ even, and let $D \geq 3$. Then the residual slice $y = -(c/2+1)e_1$ is noncentral, has remaining radii $p = t - c/2 - 1 \geq 0$ and $q = t - c/2 + 1 \leq t$, and carries*

$$\Phi_{p,q}(R, D) = \begin{cases} H, & D \geq 4, \\ \lceil H/2 \rceil, & D = 3, \end{cases}$$

which is strictly positive on every active shell.

Proof The slice has $r = c/2+1$ and $r' = c/2 - 1$, so $|r - r'| = 2$ and $q = p+2$; moreover $R \geq D \geq 3$ and $c = s - R$ even force $c \leq 2t-4$, hence $p \geq 1$. With $q = p+2$, the clipped interval satisfies $I(c') = [-p,\, p+2-c']$ for every $c' \geq 2$, while $I(1) = I(0) = [-p, p]$. Since $D \leq R \leq 2t - c = 2p+2$, the interval $I(D)$ is nonempty, and $|I(R)| = p + q - R + 1 = 2t - c - R + 1 = H$.

If $D \geq 4$, then $I(D-2)$ extends $I(D)$ by the two integers $p+3-D$ and $p+4-D$, one of each parity, so both parity brackets in (13) equal one and $\Phi_{p,q}(R, D) = n_0(p,q;R) + n_1(p,q;R) = |I(R)| = H$. If $D = 3$, then $I(1) = [-p, p]$ exceeds $I(3) = [-p, p-1]$ by the single point p, so exactly the bracket of parity $\epsilon \equiv p \pmod 2$ equals one. The interval $I(R) = [-p,\, p+2-R]$ has H points and its left endpoint $-p$ has parity ϵ, so $n_\epsilon(p,q;R) = \lceil H/2 \rceil$. □

Theorem 4.25 (Complete classification of minimizing arcs) *Let $d \geq 3$, $t \geq 1$, $4 \leq s \leq 2t$, and $H = 2t - s + 1$. Every orbit representative (a, b, w) with $a \geq b + 2$ satisfies exactly one of Cases I–IV of Lemma 4.20, so the following lists, which resolve equality case by case, are exhaustive. Modulo signed permutations, the arcs of $\mathcal{P}_d(s)$ attaining $\kappa_{d,t,s}$ are exactly the following.*

(i) *If $H \geq 3$, that is, $4 \leq s \leq 2t-2$: the balanced-gap axial family (37), and nothing else. In particular the sharp layer consists of exactly $\lfloor s/2 \rfloor - 1$ arcs, a number independent of both d and t.*

(ii) *If $H = 2$, that is, $s = 2t - 1$: the family (37), its continuation to $c = 1$ (the arc with $b = (s-3)/2$), and every arc supported on two coordinates.*

(iii) *If $H = 1$, that is, $s = 2t$: the family (37) and every arc supported on two coordinates.*

For $s \in \{2,3\}$ the shell carries one, respectively two, arcs and Theorem 4.17 already lists the minimizers; for $d = 2$ every arc of $\mathcal{P}_2(s)$ has the same weight $2t - s + 1$ by the equal-radius clause of Theorem 4.1, so the sharp layer is the whole shell.

Proof Fix an arc with data (R, D, w) and $c = s - R$ as above. The proof of Theorem 4.17 bounds its weight below by $\kappa_{d,t,s} = \sum_k S_{m-1}(k)(H - 2k)_+$; by Lemma 4.20 it suffices to decide equality in each of Cases I–IV.

Case I, family arcs: $D = 2$, $w = ce_1$, $c \geq 2$. Every noncentral slice has remaining radii with $|p - q| = |r - r'| \geq 2$ and contributes zero by Lemma 4.21. On the central slices, Lemma 4.15 turns the cumulative bound (27) into an equality at every K, hence termwise $A_k = S_{m-1}(k)$ and $\eta = 1$ when c is even, and $A_k = 2S_{m-1}(k)$ and $\eta = 1/2$ when c is odd. Either way the weight equals $\sum_k S_{m-1}(k)(H - 2k)_+ = \kappa_{d,t,s}$: every family arc is minimizing, on every active shell.

Case II, outer arcs: $c = 0$, or $(D, c) = (2, 1)$. By Lemma 4.22, their weights W_0 and W_1 equal $\kappa_{d,t,s}$ exactly when $H \leq 2$ and exceed it otherwise. Note that a gap-two arc has $c \equiv s \pmod 2$, so $c = 1$ occurs only on odd shells; this is consistent with $H = 2 \iff s = 2t - 1$ odd, and no such arc exists when $H = 1$.

Case III, spread residuals. If w has two or more nonzero coordinates (possible only for $d \geq 4$), Lemma 4.23 gives strict inequality on every active shell.

Case IV, concentrated residuals with wide gap: $w = ce_1$, $c \geq 1$, $D \geq 3$. If c is odd, every central slice has $|p - q| = 1$, but $D \geq 3$ places it in the full branch of (15), so $\eta = 1$ while (27) bounds the cumulative counts below by $2L_{m-1}(K)$; Lemma 4.16 then gives weight at least $2\kappa_{d,t,s} > \kappa_{d,t,s}$. If c is even, the central contribution equals $\kappa_{d,t,s}$ exactly as for family arcs, and Lemma 4.24 exhibits a noncentral slice of strictly positive weight.

By Lemma 4.20 the four verdicts cover every arc exactly once, and they assemble into the three lists. □

Besides the constant, Theorem 4.25 records the multiplicity of the sharp layer: it is $\lfloor s/2 \rfloor - 1$ on every interior shell and jumps on the two outermost shells, where the entire two-coordinate boundary of the shell becomes sharp. Cubes have multiplicity one (Corollary 5.2); the mixed family is generically unique, with the single $b = 1$ boundary exception of Theorem 5.4. So the degeneracy of the minimizing set is body-dependent as well.

The stability inequality (22) compares the lens deficit of an arbitrary comparable pair with $\kappa_{d,t,s}$ times its transfer distance. Theorem 4.25 decides equality at distance one; the next theorem decides it at every distance and shows that the classification of arcs already contains the complete answer. The content is a concatenation obstruction: two minimizing arcs never compose into a length-two geodesic on an interior shell, so equality in (22) cannot propagate beyond a single transfer there, while on the two outermost shells it propagates exactly along the two-coordinate boundary path. Figure 10 contrasts the two behaviours.

Theorem 4.26 (Rigidity of stability extremizers) *Let $d \geq 3$, $t \geq 1$, $4 \leq s \leq 2t$, $H = 2t - s + 1$, and let $\lambda \succeq \mu$ in $\mathcal{P}_d(s)$. Then*

$$J^{(d)}_{t,t}(\mu) - J^{(d)}_{t,t}(\lambda) = \kappa_{d,t,s}\, d_M(\lambda, \mu) \tag{38}$$

holds if and only if one of the following does:

(i) $\lambda = \mu$;

(ii) $d_M(\lambda, \mu) = 1$ *and* $\lambda \to \mu$ *is a minimizing arc of Theorem 4.25;*

(iii) $H \le 2$ *and both partitions are supported on two coordinates:* $\lambda = (a_1, s - a_1, 0^{d-2})$ *and* $\mu = (a_2, s - a_2, 0^{d-2})$ *with* $s \ge a_1 \ge a_2 \ge \lceil s/2 \rceil$.

In particular, on every interior shell, that is for $H \ge 3$, *every comparable pair with* $d_M(\lambda, \mu) \ge 2$ *obeys the strict integer gap*

$$J_{t,t}^{(d)}(\mu) - J_{t,t}^{(d)}(\lambda) \;\ge\; \kappa_{d,t,s}\, d_M(\lambda, \mu) + 1. \tag{39}$$

Proof Sufficiency. Case (i) is trivial and case (ii) is the definition of attainment. For case (iii), put $L = a_1 - a_2 = d_M(\lambda, \mu)$ and join the pair by the two-coordinate path $\nu_i = (a_1 - i,\ s - a_1 + i,\ 0^{d-2})$, $0 \le i \le L$. Each step is legal: its donor $a_1 - i \ge a_2 + 1 \ge \lceil s/2 \rceil + 1$ exceeds its recipient $s - a_1 + i$ by at least two. Each step is an arc with vanishing residual, so by Lemma 4.22 its weight is $W_0 = \kappa_{d,t,s}$ because $H \le 2$. Telescoping the lens counts along the path gives (38).

Necessity. Assume (38) with $L = d_M(\lambda, \mu) \ge 1$; the case $L = 0$ is (i). By Lemma 4.7 choose a chain $\lambda = \nu_0 \to \nu_1 \to \cdots \to \nu_L = \mu$ of exactly L arcs. Telescoping, the deficit is the sum of the L arc weights, each at least $\kappa_{d,t,s}$; equality with $\kappa_{d,t,s} L$ forces every arc of the chain to be minimizing, and $L = 1$ is case (ii). Let $L \ge 2$. Since d_M is one half of an ℓ_1 metric and consecutive chain steps have $d_M = 1$, the triangle inequality forces $d_M(\nu_i, \nu_{i+2}) = 2$ for every i: a shorter value would splice into a chain from λ to μ of fewer than L steps, contradicting Lemma 4.7.

We claim that a pair of minimizing arcs $\nu \to \nu' \to \nu''$ with $d_M(\nu, \nu'') = 2$ forces $H \le 2$ with all three partitions supported on two coordinates. By Theorem 4.25 each of the two arcs is either a member of

$$F_b:\ \operatorname{sort}(c, b+2, b, 0^{d-3}) \to \operatorname{sort}(c, b+1, b+1, 0^{d-3}), \qquad c = s - 2b - 2 \ge 1, \tag{40}$$

which covers the family (37) together with its $c = 1$ continuation and hence a superset of the minimizing gap-two arcs, or, only when $H \le 2$, an arc supported on two coordinates, $(a, s-a, 0^{d-2}) \to (a-1, s-a+1, 0^{d-2})$. We inspect the four combinations.

If both arcs are supported on two coordinates, the claim holds.

If the first arc is F_b, then $\nu' = \operatorname{sort}(c, b+1, b+1, 0^{d-3})$ has at least three nonzero entries, so the second arc cannot be supported on two coordinates and must be some $F_{b'}$ with source ν', giving the multiset identity $\{c, b+1, b+1\} = \{c', b'+2, b'\}$ with $c' = s - 2b' - 2 \ge 1$ after discarding common zeros; $b' = 0$ is impossible because its side then has only the two nonzero entries c' and 2. Since $b'+2 \ne b'$, the repeated entry forces $c' = b+1$ and $\{b'+2, b'\} = \{c, b+1\}$, leaving two possibilities. Either $b' = b+1$ and $c = b+3$, whence $\nu = (b+3, b+2, b, 0^{d-3})$ and $\nu'' = \operatorname{sort}(b+1, b+2, b+2, 0^{d-3}) = (b+2, b+2, b+1, 0^{d-3})$; or $b'+2 = b+1$ and $b' = c = b-1$, whence $\nu = (b+2, b, b-1, 0^{d-3})$ and $\nu'' = (b+1, b, b, 0^{d-3})$. In both cases the sorted difference $\nu - \nu'' = (1, 0, -1, 0^{d-3})$ gives $d_M(\nu, \nu'') = 1$, a contradiction.

If the first arc is supported on two coordinates and the second is $F_{b'}$, then $\nu' = (a-1, s-a+1, 0^{d-2})$ has exactly two nonzero entries, so as before $b' = 0$ and $\{a-1, s-a+1\} = \{s-2, 2\}$. Legality of the first arc gives $a - 1 \ge s - a + 1$, so $a - 1 = s - 2$ and $s - a + 1 = 2$, that is $a = s - 1$ (for $s = 4$ both entries equal 2 and again $a = 3$). Then $\nu = (s-1, 1, 0^{d-2})$ and $\nu'' = (s-2, 1, 1, 0^{d-3})$, so again $d_M(\nu, \nu'') = 1$, a contradiction.

Hence every window $(\nu_i, \nu_{i+1}, \nu_{i+2})$ consists of two two-coordinate arcs, so $H \le 2$ and every ν_i, in particular λ and μ, is supported on two coordinates; sorting gives the normal form in (iii), with $a_2 \ge \lceil s/2 \rceil$ because μ is a decreasing partition. Finally, for $H \ge 3$ only (i) and (ii) can occur, so a comparable pair with $d_M \ge 2$ satisfies strict inequality in (22); both sides of (38) are integers, which yields (39). □

Remark 4.27 (Distance-one rigidity) On interior shells the extremal set of the sharp inequality (22) is as small as it could possibly be: single arcs of the balanced-gap axial family, with no extremal pair at any larger distance. The mechanism is not positivity of a second-order gap but a combinatorial obstruction—the family is an antichain under concatenation: the target of a family arc is never the source of another except in two degenerate patterns, and in both the endpoints collapse to transfer distance one, so the would-be extremal chain is not a geodesic. On the two outermost shells the obstruction disappears in exactly one direction, and the extremal pairs are precisely the segments of the two-coordinate boundary path. For $s \in \{2, 3\}$ every comparable pair has $d_M \leq 1$ and the statement is vacuous.

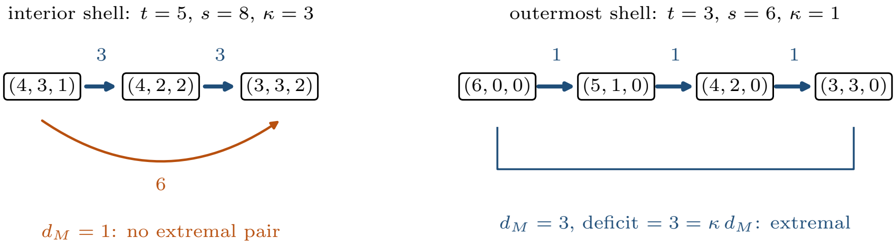

Fig. 10 The concatenation obstruction of Theorem 4.26, in $d = 3$. Left, an interior shell: the two consecutive arcs are minimizing, yet their endpoints already lie at transfer distance one, so the chain is not a geodesic and the pair is not extremal. Right, an outermost shell: the two-coordinate path is a geodesic whose every segment is minimizing, and equality in (38) persists at all distances. Arrow labels are lens deficits.

The simultaneous two-radius order of Theorem 4.4 lifts to a whole cone of Lee-radial correlations, together with an exact criterion for equality in terms of the support of the mixed differences of the profile. That lifting uses none of the sharp theory below and is deferred to Appendix A.

5 Cubes, capped cross-polytopes, and mixed sums

Corollary 3.5 fixes the sign for every convex hyperoctahedral body. It does not determine the smallest positive exposure sum. We now evaluate a second sharp family and then show two ways in which the quantitative layer changes between the cross-polytope and cube.

5.1 Integer cubes

The product factorization below is elementary; the new point is its sharp optimization over the fixed-shell transfer graph.

For $t \in \mathbb{Z}_{\geq 0}$, let

$$Q_t^d = [-t, t]^d \cap \mathbb{Z}^d.$$

For $p, q \in \mathbb{Z}_{\geq 0}$ and $k \in \mathbb{Z}_{\geq 0}$, put

$$\ell_{p,q}(k) = \begin{cases} 2\min\{p,q\}+1, & 0 \leq k \leq |p-q|, \\ p+q+1-k, & |p-q| < k \leq p+q, \\ 0, & k > p+q. \end{cases} \tag{41}$$

Theorem 5.1 (Cube product kernel and sharp shell constant) *For $u \in \mathbb{Z}^d$,*

$$g_{Q_p^d, Q_q^d}(u) = \prod_{i=1}^{d} \ell_{p,q}(|u_i|). \tag{42}$$

The sequence $\ell_{p,q}$ is nonnegative and log-concave on its support, so for $a, b \in \mathbb{Z}_{\geq 0}$ with $a \geq b+2$ the local factor satisfies

$$\ell_{p,q}(a-1)\ell_{p,q}(b+1) - \ell_{p,q}(a)\ell_{p,q}(b) \geq 0. \tag{43}$$

For equal cubes with $t \in \mathbb{Z}_{\geq 1}$, let $\kappa^{\square}_{d,t,s}$ be the least unit-transfer arc weight on the shell $\mathcal{P}_d(s)$, for $s \in \mathbb{Z}_{\geq 2}$, and put $H = 2t+1$. Then

$$\kappa^{\square}_{2,t,s} = \begin{cases} 1, & s \leq H,\ s \text{ even}, \\ 2, & s \leq H,\ s \text{ odd}, \\ 0, & s \geq H+1, \end{cases} \qquad \kappa^{\square}_{d,t,s} = \begin{cases} (H+2-s)H^{d-3}, & 2 \leq s \leq H, \\ 0, & s \geq H+1, \end{cases} \quad (d \geq 3). \tag{44}$$

Thus the active shells of the equal-radius cube are exactly $2 \leq s \leq 2t+1$, one layer beyond the Lee support bound $2t$. For $d \geq 3$ and $2 \leq s \leq H$ the minimum is attained, modulo signed permutations, by the canonical axial-residual edge of Definition 4.6,

$$\mathrm{sort}(s-2, 2, 0^{d-2}) \longrightarrow \mathrm{sort}(s-2, 1, 1, 0^{d-3}). \tag{45}$$

Consequently, for comparable $\lambda \succeq \mu$ on the shell,

$$g_{Q_t^d, Q_t^d}(\mu) - g_{Q_t^d, Q_t^d}(\lambda) \geq \kappa^{\square}_{d,t,s} d_M(\lambda, \mu),$$

and the coefficient is sharp.

Proof The one-dimensional overlap of $[-p, p] \cap \mathbb{Z}$ and its translate by $-k$ is (41); coordinate independence gives (42). For (43), suppose first that all four arguments lie in the support $[0, p+q]$. There the sequence is constant and then decreases linearly, so the adjacent ratio $\ell_{p,q}(k+1)/\ell_{p,q}(k)$ equals one on the flat part and $1 - 1/(p+q+1-k)$ on the linear part, hence is nonincreasing in k. Since $a-1 \geq b+1$,

$$\frac{\ell_{p,q}(a)}{\ell_{p,q}(a-1)} \leq \frac{\ell_{p,q}(b+1)}{\ell_{p,q}(b)},$$

and cross-multiplying the positive denominators gives (43). The zero boundary is handled by the zero extension: if $\ell_{p,q}(a) = 0$ or $\ell_{p,q}(b) = 0$, the subtracted product vanishes and the inequality is immediate, while $\ell_{p,q}(a-1) = 0$ forces $a-1 > p+q$ and hence $\ell_{p,q}(a) = 0$ as well.

For equal radii and $2 \leq s \leq H$, every coordinate of a nonnegative shell representative satisfies $u_i \leq s \leq H$, so each clipped factor equals $H - u_i \geq 0$ (a factor may vanish at $u_i = H$) and

$$g_{Q_t^d, Q_t^d}(u) = \prod_i (H - u_i).$$

A transfer on coordinates (a, b) with $D = a - b \geq 2$ has exact weight

$$(D-1) \prod_{k \neq i,j} (H - u_k), \tag{46}$$

because $(H-a+1)(H-b-1)-(H-a)(H-b) = D-1$; the formula $\ell_{t,t}(k) = H-k$ remains valid throughout $0 \leq k \leq H$, with its zero endpoint at $k = H$ supplied by the zero extension in (41), so all four factors are covered.

Assume $d \geq 3$, and let the untouched coordinates have total mass $q = s-a-b$. Repeatedly applying

$$(H-x)(H-y) \geq H(H-x-y) \qquad (x, y \geq 0),$$

which holds because the difference of the two sides is xy, concentrates this mass and gives

$$\prod_{k \neq i,j} (H-u_k) \geq H^{d-3}(H-q) = H^{d-3}(H-s+a+b).$$

Since $a + b \geq D \geq 2$ and $H - s \geq 0$,

$$(D-1)(H-s+a+b) \geq H-s+2 > 0.$$

Equality requires $D = a + b = 2$ and all untouched mass in at most one coordinate; the arc (45) has exactly this form, and its untouched coordinate $s-2 \leq H$ stays inside the support. This proves the $d \geq 3$ formula on $2 \leq s \leq H$. For $d = 2$ the residual product is empty and D has the parity of s; its least admissible value is two on even shells and three on odd shells, attained by the balanced edges $(s/2+1, s/2-1) \to (s/2, s/2)$ and $((s+3)/2, (s-3)/2) \to ((s+1)/2, (s-1)/2)$, whose coordinates stay at most H for $s \leq H$.

Finally let $s \geq H+1$. Both endpoints of the arc $(s, 0^{d-1}) \to (s-1, 1, 0^{d-2})$ leaving the axial composition se_1 have first coordinate at least H, so the clipped product (42) vanishes at both endpoints and the arc weight is zero. Every arc weight is nonnegative by Corollary 3.5, whence $\kappa^{\square}_{d,t,s} = 0$. Telescoping along a shortest unit-transfer path proves the final stability statement, and an attaining arc proves sharpness. $\square$

The equality analysis inside the proof is already complete enough to classify all minimizers, and the outcome is the exact opposite of the Lee degeneracy in Theorem 4.25.

Corollary 5.2 (Rigidity of the cube sharp layer) *Let $d \geq 3$, $t \geq 1$, and $2 \leq s \leq H = 2t+1$. Modulo signed permutations, the canonical axial-residual edge (45) is the* only *arc of $\mathcal{P}_d(s)$ attaining $\kappa^{\square}_{d,t,s}$. For $d = 2$ the unique minimizer is the balanced arc listed in the proof above.*

Proof On an active shell every untouched coordinate satisfies $u_k \leq s-a-b \leq s-2 \leq H-2$, so all factors of (46) are at least two, and the weight is the exact product $(D-1)\prod_{k \neq i,j}(H-u_k)$. The chain in the proof bounds this below by $(D-1)H^{d-3}(H-s+a+b) \geq H^{d-3}(H-s+2)$. Equality in the first step forces $u_k u_l = 0$ for every pair of untouched coordinates, that is, a concentrated residual; equality in the second step forces $D = a+b = 2$, since $D = 2 < a+b$ gives $(H-s+a+b) > (H-s+2)$ and $D \geq 3$ gives $(D-1)(H-s+a+b) \geq 2(H-s+3) > H-s+2$. The remaining data $(a, b) = (2, 0)$ and $w = (s-2)e_1$ describe exactly the arc (45). For $d = 2$ the weight is $D-1$, which is uniquely minimized by the least admissible gap of the correct parity. $\square$

Thus, on every interior Lee shell the sharp layer carries $\lfloor s/2 \rfloor - 1$ arcs, whereas the cube sharp layer is a single arc on *all* of its active shells: two solved models with the same universal transfer sign, the same attaining canonical edge, and entirely different extremal degeneracy.

5.2 Capped cross-polytopes

For integers $c, t \geq 1$ with $t \leq 2c$, define

$$P_{t,c}^{(2)} = \{x \in \mathbb{Z}^2 : \|x\|_1 \leq t,\ \|x\|_\infty \leq c\}.$$

This is the Lee ball when $t \leq c$ and the full cube $[-c, c]^2 \cap \mathbb{Z}^2$ when $t = 2c$. The next proposition tracks one distinguished atomic increment as the body interpolates from cross-polytope to cube; it does not compute a shell constant.

Proposition 5.3 (Exact tent-shaped transfer magnitude) *For the first balancing arc,*

$$g_{P_{t,c}^{(2)}, P_{t,c}^{(2)}}(1,1) - g_{P_{t,c}^{(2)}, P_{t,c}^{(2)}}(2,0) = \min\{2t-1,\ 4c-2t+1\}. \tag{47}$$

Thus the increment increases with slope two through $t = c$, has the common maximum $2c-1$ at $t = c$ and $t = c+1$, and then decreases with slope minus two; it remains positive throughout. Figure 11 plots this tent alongside the Lee and cube constants.

Proof Under $(x, y) \mapsto (r, w) = (x+y, x-y)$, the body becomes the parity-lattice points satisfying

$$|r| \leq t, \qquad |w| \leq t, \qquad |r| + |w| \leq 2c.$$

For the arc $(2,0) \to (1,1)$, the coordinate sum shift remains two and the difference shift changes from two to zero. For $|r| \leq t$, let

$$m(r) = \max\{m \in \mathbb{Z}_{\geq 0} : m \equiv r \pmod 2,\ m \leq t,\ |r| + m \leq 2c\},$$

and leave $m(r)$ undefined when the fiber is empty. The two coordinate-sum fibers are indexed by r and $r+2$, so their endpoints have the same parity. With $D = 2$, Lemma 3.3 says that the fiber exposure is one exactly when both fibers are nonempty and $m(r) = m(r+2)$. Comparing the two clipped endpoints gives

$$r \in \mathcal{R} = [-t, t-2] \cap [t-2c-1,\ 2c-t-1] \cap \mathbb{Z}. \tag{48}$$

The two integer intervals have respectively $2t-1$ and $4c-2t+1$ points. The first is contained in the second when $t \leq c$, and the second is contained in the first when $t \geq c+1$. Hence $|\mathcal{R}|$ is the right-hand side of (47). $\square$

5.3 A mixed Minkowski family

Recall $C_3 = \{x \in \mathbb{R}^3 : \|x\|_1 \leq 1\}$ and let $B_\infty^3 = [-1,1]^3$. For $b \in \mathbb{Z}_{\geq 1}$, put

$$K_b = C_3 + bB_\infty^3, \qquad A_b = K_b \cap \mathbb{Z}^3.$$

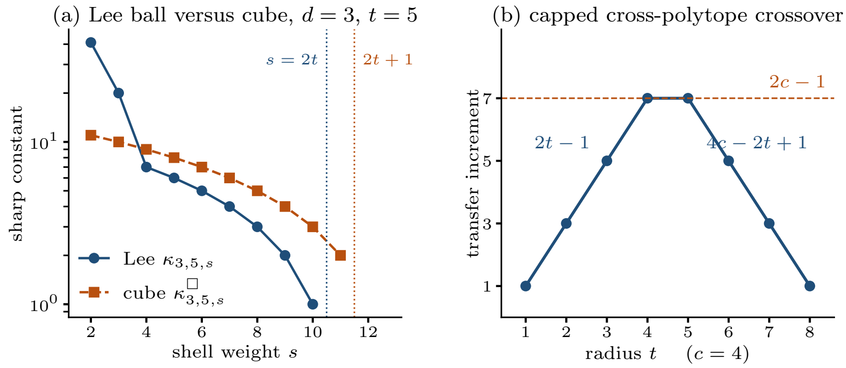

Fig. 11 Quantitative profiles of the two comparisons of this section. Panel (a) plots the Lee constant $\kappa_{3,5,s}$ of Theorem 4.17 against the cube constant $\kappa^{\square}_{3,5,s}$ of (44) on a logarithmic scale: the cube decays linearly and stays active one shell longer, while the Lee constant drops steeply off the two innermost shells. Panel (b) plots the tent of (47) for $c = 4$, with the two linear branches meeting at the plateau $2c - 1$ carried by $t = c$ and $t = c + 1$.

Equivalently,

$$A_b = \left\{ x \in \mathbb{Z}^3 : \sum_{i=1}^{3} (|x_i| - b)_+ \le 1 \right\}. \tag{49}$$

Write $N = 2b + 1$ and $S = 2(b + 1) = N + 1$.

Theorem 5.4 (Mixed outer-shell constants and a noncanonical sharp family) *For $s \in \mathbb{Z}_{\ge 2}$, put*

$$\kappa_b^{\mathrm{mix}}(s) = \min_{\lambda \to \mu \text{ in } \mathcal{P}_3(s)} \big(g_{A_b,A_b}(\mu) - g_{A_b,A_b}(\lambda) \big). \tag{50}$$

Then

$$\kappa_b^{\mathrm{mix}}(S) = 5, \qquad \kappa_b^{\mathrm{mix}}(S+1) = 4, \qquad \kappa_b^{\mathrm{mix}}(s) = 0 \quad (s \ge S + 2). \tag{51}$$

On $\mathcal{P}_3(S)$ the unique minimizing unit-transfer arc modulo signed permutations is

$$\mathrm{sort}(2b - 2, 3, 1) \longrightarrow \mathrm{sort}(2b - 2, 2, 2), \tag{52}$$

and on $\mathcal{P}_3(S + 1)$ the minimum is attained by

$$\mathrm{sort}(2b - 1, 3, 1) \longrightarrow \mathrm{sort}(2b - 1, 2, 2). \tag{53}$$

The arc (53) is the unique minimizer for $b \ge 2$; for $b = 1$ exactly one further arc, $(3, 2, 0) \to (3, 1, 1)$, also has weight four. Hence the outermost active shell of the family is $S + 1 = 2b + 3$: the sharp constant decreases from five to four on the last two active shells and then vanishes. For every b the minimizing arc on $\mathcal{P}_3(S)$ lies outside the signed-permutation orbit of the canonical axial-residual edge of Definition 4.6, and for $b \ge 2$ so does the minimizing arc on $\mathcal{P}_3(S + 1)$. In particular, this infinite family does not have the canonical axial-residual sharp edge shared by the Lee and cube formulas.

Proof Put

$$Q_b = \{-b, \ldots, b\}, \qquad E_b = \{-b - 1, b + 1\}, \qquad \mathcal{S} = \{000, 100, 010, 001\}.$$

The membership condition (49) gives the disjoint four-state decomposition

$$A_b = \bigsqcup_{\alpha \in \mathcal{S}} S_{\alpha_1} \times S_{\alpha_2} \times S_{\alpha_3}, \qquad S_0 = Q_b, \quad S_1 = E_b.$$

Let $M_k(\alpha, \beta) = |S_\alpha \cap (S_\beta - k)|$. Exact interval intersection gives

$$M_0 = \begin{pmatrix} N & 0 \\ 0 & 2 \end{pmatrix}, \qquad M_k = \begin{pmatrix} N-k & 1 \\ 1 & 0 \end{pmatrix} \quad (1 \le k \le N), \qquad M_{N+1} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \tag{54}$$

and $M_k = 0$ for $k \ge N+2$. The generic matrix remains valid up to $k = N$ inclusive, where its corner entry vanishes; only the coordinate shifts $N+1$ and beyond are exceptional, and we flag every use of the boundary matrix M_{N+1} below. Consequently

$$g_{A_b,A_b}(u) = \sum_{\alpha,\beta \in \mathcal{S}} \prod_{i=1}^{3} M_{u_i}(\alpha_i, \beta_i). \tag{55}$$

Case exhaustion. After signed permutation and sorting, every unit-transfer edge on $\mathcal{P}_3(s)$ is described by a donor coordinate, a recipient coordinate exceeded by the donor by at least two, and one untouched coordinate. Either the recipient is positive—then both transfer coordinates are positive and the untouched coordinate is positive or zero—or the recipient is zero, and the donor and the untouched coordinate carry the whole shell mass. These two cases are treated separately below on each shell and exhaust the transfer graph.

The generic transfer difference. Suppose the transfer acts on two positive coordinates $a > c \ge 1$, has gap $D = a - c \ge 2$, leaves untouched coordinate $r \ge 0$, and suppose every coordinate of u and of u' lies in $\{0, 1, \dots, N\}$. When all three shift coordinates are positive, put $A_i = N - u_i \ge 0$; expanding (55) over the sixteen state pairs reduces the state sum to

$$G(u) = A_1A_2A_3 + 2(A_1A_2 + A_1A_3 + A_2A_3) + 2(A_1 + A_2 + A_3). \tag{56}$$

If the untouched coordinate is zero, the entries $M_0(0,1) = M_0(1,0) = 0$ remove the mixed terms in that coordinate while $M_0(0,0) = N$ and $M_0(1,1) = 2$ give

$$g_{A_b,A_b}(a, c, 0) = N\big[A_aA_c + 2(A_a + A_c) + 2\big] + 2A_aA_c, \tag{57}$$

with $A_a = N - a$ and $A_c = N - c$. Direct subtraction in either case yields

$$\Delta = (N - r + 2)(D - 1), \tag{58}$$

valid on every shell on which the stated coordinate bounds hold.

The shell $S = N + 1$. If both transfer coordinates are positive, then $a \le S - c - r \le N$, so (58) applies with $N - r + 2 = a + c + 1 \ge 5$, because $a \ge c + 2$ and $c \ge 1$ force $a + c \ge 4$. Equality holds exactly when $(a, c) = (3, 1)$ and $r = S - 4 = 2b - 2$, which gives (52).

For a transfer into a zero coordinate, write the source as $(a, r, 0)$ with donor $a \ge 2$, untouched coordinate r, and $a + r = S$. For $r = 0$ the source $(N+1, 0, 0)$ uses the boundary matrix M_{N+1}, which forces the first coordinate into the state pair $(1,1)$ and gives $g_{A_b,A_b}(N+1, 0, 0) = N^2$; the target $(N, 1, 0)$ is generic. For $r \ge 1$ both endpoints are covered by (56) and (57). The same state sums give

$$\Delta = \begin{cases} N^2, & r = 0, \\ (N-1)(N+3), & r = 1, \\ a^2 + 2N - 3, & r \ge 2. \end{cases} \tag{59}$$

Every value is at least seven for $N \ge 3$. Therefore the value five in (58) is the unique minimum on $\mathcal{P}_3(S)$.

The shell $S+1=N+2$. Suppose first that both transfer coordinates are positive. If the untouched coordinate satisfies $r \geq 1$, then $a \leq N$, formula (58) applies, and now $r = N+2-a-c$ gives $N-r+2 = a+c$, so

$$\Delta = (a+c)(a-c-1) \geq 4 \cdot 1 = 4, \tag{60}$$

with equality exactly when $a+c=4$ and $a-c=2$, that is $(a,c)=(3,1)$ and $r=N-2=2b-1$; this is the arc (53). If instead $r=0$, then $a+c=N+2$. For $c \geq 2$ all coordinates stay in the generic range and (58) gives $\Delta = (N+2)(D-1) \geq N+2 \geq 5$. For $c=1$ the source is $(N+1,1,0)$: the boundary matrix M_{N+1} forces its first coordinate into the state pair $(1,1)$ and gives $g_{A_b,A_b}(N+1,1,0) = N(N-1)$, while (57) gives $g_{A_b,A_b}(N,2,0) = 2N(N-1)$; hence $\Delta = N(N-1) \geq 6$.

Next consider a transfer into a zero coordinate, with source $(a,r,0)$, donor $a \geq 2$, untouched coordinate r, and $a+r=N+2$. For $r=0$ the source $(N+2,0,0)$ has a coordinate shift of $N+2$, so its state sum vanishes, and the target value $g_{A_b,A_b}(N+1,1,0) = N(N-1)$ computed above gives $\Delta = N(N-1) \geq 6$. For $r=1$ the source is again $(N+1,1,0)$ with value $N(N-1)$, and the generic target value $g_{A_b,A_b}(N,1,1) = 2(N-1)(N+1)$ from (56) gives $\Delta = (N-1)(N+2) \geq 10$. For $2 \leq r \leq N$ both endpoints are generic; substituting $A_a = r-2$, $A_r = N-r$ into (57) and $(A_1,A_2,A_3) = (r-1, N-r, N-1)$ into (56) yields

$$\Delta = (N-r)(N-r+3) + 2(N-1) \geq 2(N-1), \tag{61}$$

with equality exactly at $r=N$, since the first summand is strictly decreasing in r. For $b \geq 2$ one has $2(N-1) \geq 8 > 4$, so (53) is the unique minimizer. For $b=1$ (so $N=3$) the value $2(N-1)=4$ is attained exactly at $r=N=3$, that is by the single further arc $(3,2,0) \to (3,1,1)$. This proves $\kappa_b^{\mathrm{mix}}(S+1) = 4$ together with the stated minimizers.

Shells $s \geq S+2$. Both endpoints of the arc $(s,0,0) \to (s-1,1,0)$ leaving the axial composition se_1 have first coordinate at least $N+2$, so $M_{u_1} = 0$ annihilates every state product and both covariogram values vanish; the arc weight is zero. Since A_b is the lattice section of a compact convex set invariant under signed coordinate permutations, Corollary 3.5 makes every arc weight nonnegative, whence $\kappa_b^{\mathrm{mix}}(s) = 0$. □

At axial radius two, the three nonnegative integer coefficient pairs in

$$\alpha C_3 + \beta B_\infty^3, \qquad \alpha + \beta = 2,$$

can be compared on the same shell $s=4$. The values and minimizing orbits are listed in Table 1: the first row follows from Theorem 5.1, the second from Theorem 5.4, and the last from the Lee values $1,2,3,4$ at $(4,0,0), (3,1,0), (2,2,0), (2,1,1)$.

Table 1 A fixed-scale quantitative transition on $\mathcal{P}_3(4)$. The transfer sign stays positive, while the sharp constant and minimizing orbit change.

body	(α,β)	κ	minimizing unit-transfer orbit(s)
$2B_\infty^3$	$(0,2)$	3	$(2,2,0) \to (2,1,1)$
$C_3 + B_\infty^3$	$(1,1)$	5	$(3,1,0) \to (2,2,0)$
$2C_3$	$(2,0)$	1	$(4,0,0) \to (3,1,0)$, $(3,1,0) \to (2,2,0)$, $(2,2,0) \to (2,1,1)$

Thus even at fixed dimension, scale, and shell, the sharp extremizer is not universal. The next section replaces this observation by an exact threshold.

6 Structural evaluation and facet threshold

In each of the three families solved above, the sharp constant is attained inside one explicit set of arcs, indexed by the recipient coordinate: the gap-two transfers whose residual mass c sits in a single coordinate. Those with $c \geq 2$ are the balanced-gap axial family of Definition 4.19, and $c \in \{0, 1\}$ is its boundary continuation. Equal Lee balls select *every* member of that family on every interior shell (Theorem 4.25); equal cubes select the single member with recipient zero, that is the canonical axial-residual edge (Corollary 5.2); and the mixed body $C_3 + bB^3_\infty$ selects the member with recipient one (Theorem 5.4). This section evaluates the mixed family in every dimension, on every shell that avoids the outer boundary layer, and shows that which member is selected is decided by one inequality between the dimension and the geometry of the body. We locate that facet threshold exactly.

Throughout, for $d \geq 1$ and $b \in \mathbb{Z}_{\geq 1}$ put

$$A_b^{(d)} = \left(C_d + bB^d_\infty\right) \cap \mathbb{Z}^d = \left\{x \in \mathbb{Z}^d : \sum_{i=1}^{d} (|x_i| - b)_+ \leq 1\right\}, \qquad g_b^{(d)} = g_{A_b^{(d)}, A_b^{(d)}},$$

so that $A_b^{(3)} = A_b$ is the body of Theorem 5.4, and write

$$N = 2b + 1, \qquad e = d - 2.$$

All shifts below are nonnegative, as we may assume after a signed permutation.

6.1 The master kernel

The four-state decomposition used for $d = 3$ collapses, in every dimension, to a single polynomial in the coordinatewise interval overlaps. Everything in this section is computed from it.

Lemma 6.1 (Master kernel) *Let $d \geq 1$, $b \geq 1$, and let $u \in \mathbb{Z}^d_{\geq 0}$ satisfy $u_i \leq N$ for every i. Put $m_i = N - u_i \geq 0$. Then*

$$g_b^{(d)}(u) = \prod_{i=1}^{d} m_i + 2 \sum_{j=1}^{d} \prod_{i \neq j} m_i + \sum_{\substack{j \neq k \\ u_j \geq 1,\ u_k \geq 1}} \prod_{i \neq j,k} m_i, \tag{62}$$

the last sum running over ordered pairs. In particular $g_b^{(d)}(u) > 0$ whenever $u_i < N$ for every i.

Proof The membership condition $\sum_i (|x_i| - b)_+ \leq 1$ says that at most one coordinate leaves the central range $Q_b = \{-b, \dots, b\}$, and that it then lies in $E_b = \{-(b+1), b+1\}$. With

$S_0 = Q_b$, $S_1 = E_b$, and $\mathcal{S} = \{\varepsilon \in \{0,1\}^d : \sum_i \varepsilon_i \le 1\}$, this is the disjoint decomposition

$$A_b^{(d)} = \bigsqcup_{\varepsilon \in \mathcal{S}} S_{\varepsilon_1} \times \cdots \times S_{\varepsilon_d}.$$

Sorting the points of $A_b^{(d)} \cap (A_b^{(d)} - u)$ by the block containing x and the block containing $x + u$ gives

$$g_b^{(d)}(u) = \sum_{\varepsilon,\eta \in \mathcal{S}} \prod_{i=1}^{d} M_{u_i}(\varepsilon_i, \eta_i), \qquad M_k(\varepsilon, \eta) = |S_\varepsilon \cap (S_\eta - k)|. \tag{63}$$

Exact interval intersection evaluates these one-dimensional kernels: for $0 \le k \le N$,

$$M_k(0,0) = N - k, \qquad M_k(0,1) = M_k(1,0) = \mathbf{1}[k \ge 1], \qquad M_k(1,1) = 2 \cdot \mathbf{1}[k = 0].$$

Indeed $b + 1 - k \in Q_b$ exactly for $1 \le k \le N$, which gives the two mixed entries, while $E_b \cap (E_b - k)$ is empty unless $k = 0$ or $k = N + 1$.

Each $\varepsilon \in \mathcal{S}$ is either 0 or a standard basis vector, so (63) has exactly four types of term. The pair $\varepsilon = \eta = 0$ contributes $\prod_i m_i$. The three types with support in a single coordinate j contribute

$$\big(M_{u_j}(1,0) + M_{u_j}(0,1) + M_{u_j}(1,1)\big) \prod_{i \ne j} m_i = 2 \prod_{i \ne j} m_i,$$

because the bracket equals 2 both when $u_j \ge 1$ and when $u_j = 0$, so one formula covers zero and positive coordinates alike. The remaining terms have $\varepsilon = e_j$, $\eta = e_k$ with $j \ne k$ and contribute $\mathbf{1}[u_j \ge 1]\mathbf{1}[u_k \ge 1] \prod_{i \ne j,k} m_i$. Summing the four types gives (62). If $u_i < N$ for every i, then every $m_i \ge 1$ and the first product is already positive. $\square$

For $e \ge 1$ and $0 \le r \le N$ write

$$\Gamma_b^{(e)}(r) = g_b^{(e)}(re_1) = (N + 2 - r)N^{e-1} + 2(e-1)(N - r)N^{e-2}, \tag{64}$$

the second term being absent when $e = 1$; this is the self-overlap of the same body *two dimensions down*, at the concentrated shift re_1. It is strictly decreasing in r on $\{0, \ldots, N\}$, with

$$\Gamma_b^{(e)}(r) - \Gamma_b^{(e)}(r+2) = 2N^{e-1} + 4(e-1)N^{e-2}. \tag{65}$$

6.2 The edge law

Every unit transfer now falls into one of exactly two regimes, according to whether the recipient coordinate is positive or zero. The first regime obeys a clean factorization; the second pays an explicit surcharge.

Lemma 6.2 (Edge law) *Let $d \ge 2$, $b \ge 1$, and let $u \in \mathbb{Z}_{\ge 0}^d$ have all coordinates at most N. Let the transfer act on a donor $a = u_1$ and a recipient $c = u_2$ with $a \ge c + 2$, put $D = a - c$, let $w = (u_3, \ldots, u_d)$ be the residual, and set $u' = u - e_1 + e_2$. Assume also $u'_1 = a - 1 \le N$ and $u'_2 = c + 1 \le N$. Put*

$$P(w) = \prod_{i=1}^{e} m_i, \qquad F(w) = \sum_{j:\, w_j \ge 1} \prod_{i \ne j} m_i, \qquad m_i = N - w_i, \tag{66}$$

with the conventions $g_b^{(0)}(0) = P = 1$ *for the empty products and* $F = 0$ *for the empty sum when* $e = 0$*. Then*

$$g_b^{(d)}(u') - g_b^{(d)}(u) = \begin{cases} (D-1)\, g_b^{(e)}(w), & c \geq 1, \\ (a-1)\, g_b^{(e)}(w) + 2P(w) + 2(N-a)F(w), & c = 0. \end{cases} \tag{67}$$

Proof Split the three sums of (62) according to how many of the two indices j, k lie in $\{1, 2\}$. Write $E_1 = \sum_j \prod_{i \neq j} m_i$ and $E_2 = \sum_{j \neq k,\, w_j w_k \geq 1} \prod_{i \neq j,k} m_i$ for the corresponding residual sums, so that $g_b^{(e)}(w) = P + 2E_1 + E_2$ by (62) in dimension e; for $e = 0$ all three residual quantities are empty, $P = 1$ and $E_1 = E_2 = F = 0$, and the identity reads $g_b^{(0)}(0) = 1$. One obtains

$$\begin{aligned} g_b^{(d)}(u) = m_1 m_2 P + 2(m_1 + m_2)P + 2m_1 m_2 E_1 \\ + 2\mathbf{1}[c \geq 1]P + 2\big(m_2 + m_1 \mathbf{1}[c \geq 1]\big)F + m_1 m_2 E_2, \end{aligned} \tag{68}$$

where we used $a \geq 2$, hence $\mathbf{1}[a \geq 1] = 1$. The transfer replaces $(m_1, m_2) = (N-a, N-c)$ by $(m_1 + 1, m_2 - 1)$; the sum $m_1 + m_2$ is preserved and

$$(m_1 + 1)(m_2 - 1) - m_1 m_2 = m_2 - m_1 - 1 = D - 1. \tag{69}$$

If $c \geq 1$, both endpoints have $\mathbf{1}[c \geq 1] = 1$, so in (68) only the coefficient $m_1 m_2$ changes, and by (69) the increment is $(D-1)(P + 2E_1 + E_2)$, which is the first line of (67).

If $c = 0$, then $m_2 = N$ at the source and the target has $(m_1', m_2') = (N-a+1, N-1)$ with both indicator terms present. Here $m_1' m_2' - m_1 m_2 = (N-a+1)(N-1) - (N-a)N = a-1$ and $m_1' + m_2' = 2N - a = m_1 + m_2$. Subtracting the two instances of (68) therefore gives

$$(a-1)P + 2(a-1)E_1 + (a-1)E_2 + 2P + \big[2(2N-a) - 2N\big]F = (a-1)g_b^{(e)}(w) + 2P + 2(N-a)F,$$

which is the second line. □

The surcharge $2P + 2(N-a)F$ carried by a transfer into a zero coordinate is a facet effect. It arises from the single entry $M_0(1,1) = 2$, that is from the two extreme layers $x_i = \pm(b+1)$ of the cube summand overlapping themselves; a coordinate is shifted by zero only at a recipient that is zero, so no other arc sees it. This is how the canonical axial-residual edge, the cheapest arc for the cube, is penalized once a cross-polytope summand is present.

We also need the elementary fact that a concentrated residual is the cheapest one, simultaneously for all three functionals appearing in (67).

Lemma 6.3 (Residual monotonicity) *Let* $e \geq 1$ *and let* $\lambda \succeq \mu$ *be distinct nonnegative compositions of* r *into* e *parts with* $r \leq N-2$*. Then*

$$g_b^{(e)}(\mu) > g_b^{(e)}(\lambda), \qquad P(\mu) \geq P(\lambda), \qquad F(\mu) \geq F(\lambda).$$

Consequently, among all $w \in \mathbb{Z}_{\geq 0}^e$ *with* $\|w\|_1 = r \leq N-2$ *the concentrated residual* re_1 *minimizes* $g_b^{(e)}$*,* P*, and* F *simultaneously, and it is the unique minimizer of* $g_b^{(e)}$*.*

Proof By the majorization theorem of Hardy–Littlewood–Pólya, μ is obtained from λ by a finite chain of unit balancing transfers, so it suffices to treat one transfer; since re_1 majorizes every composition of r into e parts, the three inequalities then give the last statement. A transfer needs $e \geq 2$. All coordinates involved are at most $r \leq N-2$, so every $m_i \geq 2$ and Lemma 6.2 applies in dimension e, where the residual has dimension $e-2 \geq 0$. Its two cases give increments $(D-1)g_b^{(e-2)}(w) \geq g_b^{(e-2)}(w) > 0$ and $(a-1)g_b^{(e-2)}(w)+2P+2(N-a)F > 0$ respectively, using Lemma 6.1 for positivity and, when $e=2$, the conventions of Lemma 6.2. This gives the strict statement for $g_b^{(e)}$.

For P, the transfer multiplies the two active factors and leaves the rest fixed, so P increases by $(D-1)\prod_{i\geq 3} m_i \geq 0$ when the recipient is positive, and by $(a-1)\prod_{i\geq 3} m_i \geq 0$ when it is zero, by (69) and the computation in the proof of Lemma 6.2. For F, write $Q = \prod_{i\geq 3} m_i$ and $G = \sum_{j\geq 3,\, w_j\geq 1}\prod_{i\geq 3,\, i\neq j} m_i$. If the recipient is positive, then $F = (m_1+m_2)Q + m_1 m_2 G$ and the increment is $(D-1)G \geq 0$. If the recipient is zero, then $F = NQ + (N-a)NG$ at the source and $(2N-a)Q + (N-a+1)(N-1)G$ at the target, so the increment is $(N-a)Q + (a-1)G \geq 0$. □

6.3 The structural theorem

Theorem 6.4 (Structural evaluation and facet threshold) *Let $d \geq 3$ and $b \geq 2$, put $N = 2b+1$ and $e = d-2$, and let s satisfy $4 \leq s \leq N$. Let*

$$\kappa_b^{(d)}(s) = \min_{\lambda\to\mu \text{ in } \mathcal{P}_d(s)} \big(g_b^{(d)}(\mu) - g_b^{(d)}(\lambda)\big)$$

be the least unit-transfer weight on the shell. Then $\kappa_b^{(d)}(s)$ is the smaller of the two values

$$\Gamma_b^{(e)}(s-4) \qquad \text{and} \qquad \Gamma_b^{(e)}(s-2) + 2N^{e-1}(2N-s), \tag{70}$$

the first being the weight of the balanced-gap arc with recipient one,

$$\operatorname{sort}(s-4,3,1,0^{d-3}) \longrightarrow \operatorname{sort}(s-4,2,2,0^{d-3}), \tag{71}$$

and the second the weight of the canonical axial-residual edge of Definition 4.6. Their difference is governed by the single quantity

$$\Theta_{d,b}(s) = N(2N-s-1) - 2(d-3), \tag{72}$$

namely

$$\Big[\Gamma_b^{(e)}(s-2) + 2N^{e-1}(2N-s)\Big] - \Gamma_b^{(e)}(s-4) = 2N^{e-1}(2N-s-1) - 4(d-3)N^{e-2}, \tag{73}$$

which has the sign of $\Theta_{d,b}(s)$. Consequently, modulo signed permutations:

1. *if $\Theta_{d,b}(s) > 0$, then $\kappa_b^{(d)}(s) = \Gamma_b^{(e)}(s-4)$ and the arc (71) is the unique minimizer;*
2. *if $\Theta_{d,b}(s) < 0$, then $\kappa_b^{(d)}(s) = \Gamma_b^{(e)}(s-2)+2N^{e-1}(2N-s)$ and the canonical axial-residual edge is the unique minimizer;*
3. *if $\Theta_{d,b}(s) = 0$, the minimum is attained exactly on these two orbits.*

In particular every such shell is active, and for comparable $\lambda \succeq \mu$ on it, $g_b^{(d)}(\mu) - g_b^{(d)}(\lambda) \geq \kappa_b^{(d)}(s)\, d_M(\lambda,\mu)$ with a sharp constant.

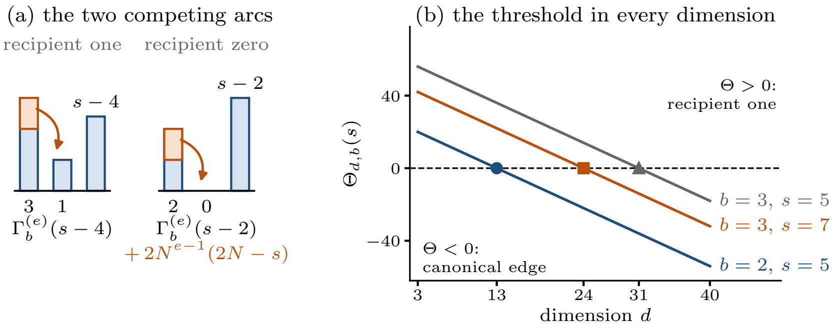

Fig. 12 The competition of Theorem 6.4. In (a) the two candidate arcs of (70), drawn as donor, recipient, and residual coordinates with the transferred unit shaded; the recipient-zero arc buys a larger residual and pays the facet surcharge. In (b) the threshold (72) against the dimension for three pairs (b,s): it decreases with slope -2, and the marked integer zeros $d=13,24,31$ are the smallest ties, where both arcs are minimizing.

Proof Since $s\le N$, every nonnegative representative of a vertex of $\mathcal{P}_d(s)$ has all coordinates at most $s\le N$, so Lemmas 6.1 and 6.2 apply to every arc of the shell, and the two lines of (67) exhaust the arcs.

Arcs with positive recipient. Here $a\ge c+2$ and $c\ge 1$ force $a+c\ge 4$, so the residual mass is $\|w\|_1=s-a-c\le s-4\le N-4$. By Lemma 6.3 and the strict monotonicity (65),

$$(D-1)g_b^{(e)}(w)\ \ge\ g_b^{(e)}(w)\ \ge\ \Gamma_b^{(e)}(\|w\|_1)\ \ge\ \Gamma_b^{(e)}(s-4),$$

and equality throughout forces $D=2$, $a+c=4$, hence $(a,c)=(3,1)$, together with a concentrated residual: exactly the arc (71), whose weight is therefore $\Gamma_b^{(e)}(s-4)$.

Arcs with zero recipient. Here $D=a$ and $\|w\|_1=s-a\le s-2\le N-2$; assume first $a\le s-1$, so that $w\ne 0$. By Lemma 6.3 all three functionals in the second line of (67) are minimized, for fixed a, by the concentrated residual $w=(s-a)e_1$, and strictly so for $g_b^{(e)}$ unless w is concentrated. For that residual $P=(N-s+a)N^{e-1}$ and $F=N^{e-1}$, the latter because exactly one residual coordinate is nonzero, so the surcharge is

$$2P+2(N-a)F=2N^{e-1}\big[(N-s+a)+(N-a)\big]=2N^{e-1}(2N-s),$$

independent of a. Hence the weight equals $(a-1)\Gamma_b^{(e)}(s-a)+2N^{e-1}(2N-s)$, and since both $a-1$ and $\Gamma_b^{(e)}(s-a)$ are positive and strictly increasing in a, the minimum over $2\le a\le s-1$ is attained exactly at $a=2$ with a concentrated residual, that is exactly at the canonical axial-residual edge, with value $\Gamma_b^{(e)}(s-2)+2N^{e-1}(2N-s)$.

The endpoint $a=s$. Then $w=0$, so F is an empty sum, $P=N^e$, and the weight is $(s-1)\Gamma_b^{(e)}(0)+2N^e$ instead of the surcharge form above. Since $s\ge 4$ and $\Gamma_b^{(e)}$ is positive and strictly decreasing, $(s-1)\Gamma_b^{(e)}(0)-\Gamma_b^{(e)}(s-2)>2\Gamma_b^{(e)}(0)\ge 2(N+2)N^{e-1}$ by (64), so this weight exceeds $\Gamma_b^{(e)}(s-2)+2N^{e-1}(2N-s)$ by more than $2(N+2)N^{e-1}+2N^e-2N^{e-1}(2N-s)=2N^{e-1}(s+2)>0$. Hence the canonical axial-residual edge minimizes the whole zero-recipient regime.

Comparison. Formula (73) is (65) with $r=s-4$, and since $N^{e-2}>0$ its right-hand side has the sign of $N(2N-s-1)-2(e-1)=\Theta_{d,b}(s)$, which for $e=1$ reduces to $N(2N-s-1)>0$. The three cases follow, and each candidate value is positive because $s\le N$, so the shell is active.

For the last statement let $\lambda \succeq \mu$ on the shell. By Lemma 4.7 they are joined by a chain of $d_M(\lambda,\mu)$ unit balancing transfers, each an arc of the shell. Every arc has $g_b^{(d)}$-increment at least $\kappa_b^{(d)}(s)$, so summing along the chain gives $g_b^{(d)}(\mu) - g_b^{(d)}(\lambda) \geq \kappa_b^{(d)}(s)\, d_M(\lambda,\mu)$; an arc attaining the minimum is a comparable pair at distance one for which this is an equality, so no larger constant works. □

Two features of Theorem 6.4 deserve separate mention.

First, both candidates in (70) are built from self-overlaps of the same body two dimensions down: the first is such a count outright, the second is one plus the facet surcharge. Both previous evaluations have this shape: the cube constant (44) is $(H+2-s)H^{d-3} = g_{Q_t^{d-2},Q_t^{d-2}}\big((s-2)e_1\big)$ and the first candidate is $g_b^{(d-2)}\big((s-4)e_1\big)$, because by Lemma 6.2 and its cube counterpart (46) an arc with positive recipient weighs $(D-1)$ times the self-overlap of the body in dimension $d-2$ at the residual shift. The sharp layer is then a $(d-2)$-dimensional enumeration plus the universal sign of Corollary 3.5 in that dimension, as used in Lemma 6.3. For the Lee ball the factorization fails on *every* arc: the increment is the residual convolution of Proposition 4.2, not a product, and the resulting degeneracy is the family of Theorem 4.25.

Remark (Three regimes of one family) Index the gap-two arcs with concentrated residual by the recipient coordinate. On interior shells equal Lee balls select every member of residual mass at least two, the whole family of Definition 4.19; equal cubes select recipient zero; and $C_d + bB_\infty^d$ selects recipient one when $\Theta_{d,b}(s) > 0$ and recipient zero when $\Theta_{d,b}(s) < 0$. The surcharge $2N^{e-1}(2N-s)$ paid by the recipient-zero arc is of order N^e and comes from the two capped layers of the cube summand; what it buys is the larger residual mass $s-2$ instead of $s-4$, worth $2N^{e-1} + 4(d-3)N^{e-2}$ by (65). Balancing the two gives (72): the facet effect dominates in low dimension, the $d-3$ free residual coordinates once $2(d-3)$ exceeds $N(2N-s-1)$. Both signs occur, and so does the tie: since N is odd, $\Theta_{d,b}(s) = 0$ has solutions exactly for odd s, at $d = 3 + N(2N-s-1)/2$; the smallest are $(d,b,s) = (13,2,5)$, $(24,3,7)$, $(31,3,5)$, and $(39,4,9)$. Figure 12 shows the two arcs and the sign change. For $d = 3$ we have $\Theta_{3,b}(s) = N(2N-s-1) > 0$ on the whole range, so the recipient-one arc always wins and (64) gives the closed form $\kappa_b^{(3)}(s) = N + 6 - s$.

Remark (Relation to Theorem 5.4 and scope) For $d = 3$ the two evaluations complete each other. The closed form $\kappa_b^{(3)}(s) = N + 6 - s$ covers $4 \leq s \leq N$, and Theorem 5.4 evaluates the two remaining active shells $s = N+1$ and $s = N+2$, where a coordinate can reach the exceptional shift $N+1$ and the master kernel acquires a boundary correction; its values five and four are exactly $N + 6 - s$ there. So for $d = 3$ the constant is $N + 6 - s$ on every active shell, with the recipient-one arc minimizing except for the single additional arc at $b = 1$, $s = N+2$. For $d \geq 4$ the hypothesis $s \leq N$ confines every coordinate to the range where Lemma 6.1 is exact; the outer shells are not treated here.

7 Finite-quotient boundary

The integer-lattice hypothesis cannot be replaced by a naive finite-q analogue.

Proposition 7.1 (Wrap-around order reversal) *In $\mathbb{Z}_6^2$, equip each coordinate with cyclic Lee weight $|z|_6 = \min\{z, 6-z\}$ for $z \in \{0, \dots, 5\}$, and take radii $p = 1$ and $q = 3$. Although $(3,1) \succeq (2,2)$, one has*

$$J_{1,3}^{(2),\mathbb{Z}_6}(3,1) = 3 > 2 = J_{1,3}^{(2),\mathbb{Z}_6}(2,2). \tag{74}$$

Proof The accepted point sets inside the radius-one ball are

$$\{(0,5),(1,0),(5,0)\} \qquad \text{and} \qquad \{(0,5),(5,0)\},$$

for the two shifts, respectively. Their cardinalities prove (74). □

8 Scope and limitations

The exact transfer theorem is for finite lattice sets with centered exchange fibers, and the fixed-shell order additionally requires signed-permutation invariance. Lattice sections of compact convex sets invariant under signed coordinate permutations satisfy both and form a broad sufficient class: cross-polytopes, cubes, convex symmetric-gauge balls, their intersections, and their Minkowski sums. We do not claim the property is necessary; the examples in 3.8 show only that dropping convexity or the symmetry makes a sign reversal possible, not that every set outside the class reverses.

The Lee sharp constant is closed for every $d \geq 2$, while the arc classification and the rigidity need $d \geq 3$ and $4 \leq s \leq 2t$, with $s \in \{2,3\}$ separate and $d = 2$ uniform. For $C_d + bB_\infty^d$, Theorem 6.4 needs $4 \leq s \leq 2b+1$; the outer shells $s > 2b+1$ in dimension $d \geq 4$, where the master kernel of Lemma 6.1 acquires boundary corrections, are not treated, and Theorem 5.4 settles them only for $d = 3$. The threshold $\Theta_{d,b}(s)$ is asserted for this one-parameter family, not for $\alpha C_d + \beta B_\infty^d$ with $\alpha \geq 2$, where Proposition 5.3 evaluates one distinguished atomic arc only. The symbolic Lee evaluator of Appendix C is output-sensitive and pseudopolynomial in the radii; no bound polynomial in the dimension and the binary-encoded radius is asserted.

The ambient group is the signed-permutation group of the standard integer lattice; we claim no analogue for an arbitrary lattice basis, and no finite-q theorem, since (74) reverses the majorization order. The rejection sampling of Remark A.3 is a potential application only; its probabilistic assumptions are stated in Online Resource 1 and used nowhere here.

9 Conclusion

For every pair of finite centered exchange-fiber sets, one coordinate balancing move decomposes exactly into parity-fiber endpoint exposures, each equal to zero or one; convex hyperoctahedral lattice sections automatically have these fibers. It yields a fiberwise equality certificate along the fixed-shell transfer graph and, for equal Lee balls with $d \geq 3$, the closed sharp constant on every active shell, all minimizing arcs for $4 \leq s \leq 2t$, and the distance-one rigidity of the extremizers on interior shells. Cubes factor into interval overlaps, with a product-form constant. Section 6 locates all three behaviours inside one mechanism: for $C_d + bB_\infty^d$, an arc with positive recipient weighs its gap minus one times the self-overlap of the same body two dimensions down, while an arc into a zero coordinate pays a surcharge from the two capped layers of the cube summand. The sharp constant is therefore determined by lower-dimensional lattice-point counts, with an explicit facet surcharge for transfers into a zero coordinate, and which arc realizes it is decided by the sign of $\Theta_{d,b}(s)$. Cubes have no surcharge and Lee balls no factorization, which is why their sharp layers have one and $\lfloor s/2 \rfloor - 1$ minimizing arcs: the sign is a matter of symmetry, the sharp scale and the extremizing geometry are not.

Three problems remain open. First, the centered exchange-fiber property is sufficient for the transfer sign but not shown to be necessary; a characterization of fixed-shell Schur monotonicity by a fiber or exchange condition would be stronger. Second, the master kernel of Lemma 6.1 is special to one cross-polytope summand: $\alpha C_d + \beta B_\infty^d$ has $\alpha + 1$ fiber states, and the exposure functional replacing the surcharge there would give a two-parameter threshold surface. Third, the outer shells $s > 2b + 1$ with $d \geq 4$, where the kernel acquires boundary corrections, remain open.

Statements and Declarations

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Author contributions

The sole author designed the study, proved the results, wrote the verification code, and prepared the manuscript.

Supplementary information

The article is accompanied by a single supplementary file, Online Resource 1, containing the probabilistic material behind Remark A.3. The reproducibility package of Appendix D is not a supplementary file.

Data availability

Every numerical value in the tables is regenerated from the source code in the reproducibility package, https://github.com/ShunqiLu/covariogram-balancing, which also supplies the exact fractions and metadata as CSV, JSON, and Markdown.

Code availability

The same repository contains the Python source, tests, PowerShell and POSIX wrappers, LaTeX sources, the figure generator, and a SHA-256 manifest; no proprietary software or GPU is required.

Appendix A The Lee-radial correlation cone

The simultaneous two-radius order extends to the Lee-radial correlation cone

$$\mathcal{C}_{\mathrm{Lee}} = \{H : \mathbb{Z}^2_{\geq 0} \to \mathbb{R}_{\geq 0} : \mathrm{supp}(H) \text{ is finite and } \Delta_{12}H \geq 0\}.$$

Theorem A.1 (Order and equality on the Lee-radial correlation cone) *Let $H \in \mathcal{C}_{\mathrm{Lee}}$, with mixed forward differences*

$$\Delta_{12}H(p,q) = H(p,q) - H(p+1,q) - H(p,q+1) + H(p+1,q+1) \geq 0.$$

Define

$$K_H(u) = \sum_{x \in \mathbb{Z}^d} H(\|x\|_1, \|x+u\|_1).$$

If $\lambda, \mu \in \mathbb{Z}^d_{\geq 0}$ have equal coordinate sum and $\lambda \succeq \mu$, then

$$K_H(\lambda) \leq K_H(\mu). \tag{A1}$$

Moreover, define the lens-equality set

$$\mathcal{E}(\lambda,\mu) = \{(p,q) : J^{(d)}_{p,q}(\lambda) = J^{(d)}_{p,q}(\mu)\}.$$

Then the following equality classification is exact:

$$K_H(\lambda) = K_H(\mu) \quad \Longleftrightarrow \quad \mathrm{supp}(\Delta_{12}H) \subseteq \mathcal{E}(\lambda,\mu). \tag{A2}$$

In particular, this applies to $H(r,s) = f(r)g(s)$ for any two nonnegative nonincreasing profiles f, g of finite support, and by monotone convergence whenever the radial lifts $x \mapsto f(\|x\|_1)$ and $x \mapsto g(\|x\|_1)$ are summable on $\mathbb{Z}^d$. Thus, at fixed shift weight s, the concentrated (axial) shift se_1 minimizes, and the balanced shift—the unique minimal element of the majorization order on the shell modulo permutations, whose coordinates all equal $\lfloor s/d \rfloor$ or $\lceil s/d \rceil$—maximizes every correlation in the cone.

More precisely, for product profiles the inequality is strict for nonpermutation shifts $\lambda \succeq \mu$ of common weight s whenever there is an integer $r \geq \lceil s/2 \rceil$ such that

$$f(r) > f(r+1) \quad \text{and} \quad g(r) > g(r+1). \tag{A3}$$

In particular, for every $0 < \sigma, \tau < 1$ the correlation

$$C_{\sigma,\tau}(u) = \sum_{x \in \mathbb{Z}^d} \sigma^{\|x\|_1} \tau^{\|x+u\|_1} \tag{A4}$$

is strictly Schur-concave on each fixed-ℓ_1-weight class, modulo signed coordinate permutations.

Proof Because H has finite support and every Lee shell is finite, all sums in the finite-support argument are finite. Double telescoping gives the lower-orthant decomposition

$$H(r,s) = \sum_{p \geq r, q \geq s} \Delta_{12}H(p,q).$$

Changing the order of summation therefore yields

$$K_H(u) = \sum_{p,q \geq 0} \Delta_{12}H(p,q) J^{(d)}_{p,q}(u).$$

Theorem 4.4 applies term by term. Every summand in the difference

$$K_H(\mu) - K_H(\lambda) = \sum_{p,q\geq 0} \Delta_{12}H(p,q)\big(J^{(d)}_{p,q}(\mu) - J^{(d)}_{p,q}(\lambda)\big)$$

is nonnegative. Its sum is zero exactly when every positive coefficient multiplies a zero lens gap, proving (A2) for finitely supported profiles.

For product profiles with infinite support, define the explicit truncations

$$f_N(r) = f(r)\,\mathbf{1}_{\{r\leq N\}}, \qquad g_N(r) = g(r)\,\mathbf{1}_{\{r\leq N\}} \qquad (N \in \mathbb{Z}_{\geq 0}).$$

Each truncation is nonnegative and nonincreasing, and $H_N = f_N g_N$ has finite support with

$$\Delta_{12}H_N(p,q) = (f_N(p) - f_N(p+1))(g_N(q) - g_N(q+1)) \geq 0,$$

so the finite-support case applies to every H_N. As $N \to \infty$ the truncations increase pointwise to f and g, hence $K_{H_N}(u) \uparrow K_H(u)$ by monotone convergence when the radial lifts are summable, and the weak inequality passes to the limit; Tonelli's theorem justifies the limiting lower-orthant decomposition, whose resulting nonnegative series is zero exactly when each positive mixed-difference coefficient selects a zero lens gap. Thus the equality criterion also persists.

For strictness under (A3), fix an integer $r \geq \lceil s/2\rceil$ with $f(r) > f(r+1)$ and $g(r) > g(r+1)$. Every truncated decomposition with $N > r+1$ contains the fixed term

$$\big(f(r) - f(r+1)\big)\big(g(r) - g(r+1)\big)\big(J^{(d)}_{r,r}(\mu) - J^{(d)}_{r,r}(\lambda)\big),$$

which is strictly positive by the strict clause of Theorem 4.4, since $s \leq 2r$. All other terms are nonnegative, so the difference $K_H(\mu) - K_H(\lambda)$ is bounded below by this one positive quantity uniformly in the truncation and remains strictly positive in the limit. Taking $f(r) = \sigma^r$ and $g(r) = \tau^r$ satisfies the summability and strict-drop conditions at every radius. $\square$

Corollary A.2 (Axial minimizers of Lee-radial correlations) *Let $f : \mathbb{Z}_{\geq 0} \to \mathbb{R}_{\geq 0}$ be nonincreasing with $0 < Z_f = \sum_{x\in\mathbb{Z}^d} f(\|x\|_1) < \infty$, and let $g : \mathbb{Z}_{\geq 0} \to [0,1]$ be nonincreasing. Define the normalized Lee-radial correlation*

$$A_{f,g}(u) = \frac{1}{Z_f}\sum_{x\in\mathbb{Z}^d} f(\|x\|_1)g(\|x+u\|_1).$$

If Y has mass $f(\|y\|_1)/Z_f$, then $A_{f,g}(u)$ is the probability that a radial test with profile g accepts $Y+u$. For every finite nonempty set $R \subset \mathbb{Z}_{\geq 0}$,

$$\min_{\|u\|_1\in R} A_{f,g}(u) = \min_{r\in R} A_{f,g}(re_1). \tag{A5}$$

Consequently, the minimum of $A_{f,g}$ over $\{u : \|u\|_1 \leq s\}$ is determined by the $s+1$ axial values at $0, e_1, \ldots, se_1$. For a random shift U independent of Y with $\Pr(\|U\|_1 \in R) = 1$,

$$\Pr[\text{accept}] = \mathbb{E}[A_{f,g}(U)] \geq \min_{r\in R} A_{f,g}(re_1).$$

On a fixed shell of weight s, if there is an integer $r \geq \lceil s/2\rceil$ with $f(r) > f(r+1)$ and $g(r) > g(r+1)$, then the axial signed-permutation orbit is the unique minimizing orbit.

Proof For a fixed r, the vector re_1 majorizes the absolute-coordinate vector of every u with $\|u\|_1 = r$. The radial lift of g need not be summable, so apply Theorem A.1 to the truncated product profiles $H_N(p,q) = f_N(p)g_N(q)$ of its proof and pass to the limit using

$$0 \leq f(\|x\|_1)g(\|x+u\|_1) \leq f(\|x\|_1), \qquad \sum_{x\in\mathbb{Z}^d} f(\|x\|_1) = Z_f < \infty,$$

which dominates all truncations simultaneously. Division by Z_f gives $A_{f,g}(re_1) \le A_{f,g}(u)$ on each shell, with equality attained at re_1. Taking the minimum over $r \in R$ proves (A5); averaging proves the random-shift bound.

For the strict statement, let u have weight s with $|u|^{\downarrow}$ outside the axial orbit; then se_1 strictly majorizes $|u|^{\downarrow}$ and the two are not permutations, so there is no incomparability obstruction on this shell. The fixed-positive-term argument in the proof of Theorem A.1, applied to the gap $Z_f\big(A_{f,g}(u) - A_{f,g}(se_1)\big)$, bounds every truncated gap with $N > r+1$ below by the same positive quantity, and the dominated limit above preserves the bound. □

Remark A.3 (A potential application) When a nonincreasing Lee-radial acceptance profile is applied after a shift from prescribed Lee shells—for example in uniform-polytope rejection sampling [66] or in the lattice Fiat–Shamir-with-aborts line [67–70]—equation (A5) replaces the worst-case search over coordinate compositions by one axial evaluation per Lee weight. The probabilistic interface needed to turn this into acceptance-rate statements is developed in Online Resource 1 and is not used anywhere in this paper.

Appendix B The exact lens enumerator

For the equal-radius translated lens, write

$$N_{C_d}(t,u) = |B_1^d(t) \cap (B_1^d(t) - u)|,$$

and for $a \in \mathbb{Z}_{\ge 0}$ define the formal bivariate series

$$F_a(X,Y) = \sum_{r \in \mathbb{Z}} X^{|r|} Y^{|r+a|}.$$

For a negative coordinate shift, substituting $r \mapsto -r$ shows $F_{-a} = F_a$; thus only $|u_i|$ matters in each coordinate.

Theorem B.1 (Rational series and all-scale formula) *Let*

$$p_0(X,Y) = 1 + XY$$

and, for $a \ge 1$,

$$p_a(X,Y) = X^a + Y^a + (1 - XY) \sum_{j=1}^{a-1} X^j Y^{a-j}. \tag{B6}$$

Then

$$F_a(X,Y) = \frac{p_a(X,Y)}{1 - XY}.$$

For $u \in \mathbb{Z}^d$, put

$$P_u(X,Y) = \prod_{i=1}^{d} p_{|u_i|}(X,Y) = \sum_{i,j \ge 0} c_{ij} X^i Y^j.$$

The simultaneous norm enumerator and the lens count are

$$\sum_{x \in \mathbb{Z}^d} X^{\|x\|_1} Y^{\|x+u\|_1} = \frac{P_u(X,Y)}{(1 - XY)^d}, \tag{B7}$$

$$N_{C_d}(t,u) = \sum_{i,j\ge 0} c_{ij} \binom{d+t-\max(i,j)}{d}_+ . \tag{B8}$$

Both quantities depend on the shift through the multiset $\{|u_1|,\dots,|u_d|\}$; no coarser scalar such as $\|u\|_1$ is asserted to determine them. Here

$$\binom{d+q}{d}_+ = \begin{cases} \binom{d+q}{d}, & q\ge 0,\\ 0, & q<0. \end{cases}$$

The coefficients c_{ij} may be negative, but the sum in (B8) is a nonnegative integer. For fixed d and u, the formula is an ordinary polynomial in t after $t \ge \max\{i,j : c_{ij}\ne 0\}$.

As checks and useful special cases, equation (9) and Theorem B.1 give

$$N_{C_d}(t,2e_1) = L_d(t-1), \qquad N_{C_d}(t,e_1+e_2) = L_d(t) - L_{d-1}(t) \quad (d\ge 2, t\ge 1).$$

Proof of Theorem B.1 For $a=0$, the term $r=0$ and the two tails give

$$F_0 = 1 + 2\sum_{k\ge 1}(XY)^k = \frac{1+XY}{1-XY}.$$

For $a\ge 1$, split the integer line into $r\ge 0$, $r\le -a$, and $-a<r<0$. The first two ranges give respectively $Y^a/(1-XY)$ and $X^a/(1-XY)$; the $a-1$ interior integers give $\sum_{j=1}^{a-1} X^jY^{a-j}$. Multiplying by $1-XY$ yields (B6). The negative term contributed by $-XY\sum_j X^jY^{a-j}$ explains why the numerator need not have nonnegative coefficients.

The d coordinates are independent as formal series, so multiplication gives (B7). Expanding

$$(1-XY)^{-d} = \sum_{k\ge 0}\binom{d+k-1}{d-1} X^kY^k$$

and summing all coefficients with both exponents at most t, a numerator monomial X^iY^j allows $0\le k\le t-\max(i,j)$. The hockey-stick identity gives

$$\sum_{k=0}^{q}\binom{d+k-1}{d-1} = \binom{d+q}{d},$$

which proves (B8). For sufficiently large t every truncation branch is active and each summand is polynomial in t. □

Appendix C Exact coefficient extraction and complexity

The direct bivariate dynamic program keeps states (r,s) with $0\le r,s\le t$. For one coordinate it groups all integers x_i by $(|x_i|,|x_i+u_i|)$, multiplies the current table by these terms, and truncates both degrees. With a straightforward sparse loop there are $O(t^2)$ states and $O(t)$ relevant coordinate terms, giving $O(dt^3)$ integer additions/multiplications and $O(t^2)$ stored integers. The count never exceeds $(2t+1)^d$, so coefficient bit length is $O(d\log(t+1))$; a bit-complexity bound multiplies the operation count by the cost of integers of that length.

The dynamic program is pseudopolynomial when t is encoded in binary, not polynomial in $\log t$. The closed formula is efficient for fixed d and shift pattern, but sparse expansion of P_u may grow with the product of the coordinate numerator sizes. The artifact therefore reports both asymptotic arithmetic complexity and measured time and heap use.

Appendix D Reproducibility package

The reproducibility package is publicly available at https://github.com/ShunqiLu/covariogram-balancing, distinct from Online Resource 1, the only supplementary file submitted with the article. The repository contains the Python source, raw CSV/JSON/Markdown outputs, the LaTeX sources, the script that regenerates every figure of this paper as a vector PDF from the same exact computations, and a field-level data dictionary. Every theorem was cross-checked by an automated test suite whose exact ranges and output counts are archived in `results/quantitative_lee_rearrangement.md`; these finite computations were used during discovery and remain regression tests, and none of them enters a proof. For Section 6 the checks archived in `results/structural_threshold_scan.md` validate the master kernel (62) against direct enumeration and both cases of the edge law (67), the latter for $2 \le d \le 7$ and $1 \le b \le 4$, so including the empty-residual conventions at $d = 2$. A third check covers the donor-heavy endpoint $a = s$ of the zero-recipient regime, where the residual is empty and the surcharge takes the different value $2N^e$. They also validate the conclusion of Theorem 6.4—value, minimizing orbit, and the sign of (72)—for $2 \le b \le 6$, $3 \le d \le 39$, and every shell $4 \le s \le 2b + 1$, including the four listed threshold ties. The SHA-256 manifest covers every tracked source, result, manuscript, and external input. The external inputs specific to the probabilistic application, together with their retrieval metadata and hashes, are documented in the reproducibility notes of Online Resource 1.

The cross-platform driver regenerates all experiments, figures, tables, and the manuscript. The single timing experiment fixes seed 20260802; theorem and table data are deterministic. From the repository root, run

```
python scripts/reproduce.py --quick
python scripts/reproduce.py --full
```

The full mode requires `pdflatex`, `bibtex`, and `pdftops` on the path. The driver stops on the first failing command, prints the complete wall time, and verifies the newly generated hashes. Exact environment metadata are recorded in `results/performance_environment.json`.

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